Chapter 4

P-N Junctions

The p-n junction is one of the basic building blocks in integrated circuit. Such a junction can be formed by selective diffusion or ion implantation of n-type (or p-type) dopants into a p-type (or n-type) semiconductor sample. It has already been mentioned in Chapter 2, that the resulting dopant distribution will follow a complementary error function (erfc) or a Gaussian profile as shown in Fig. 4.1(a). However, in order to obtain simplified analytical expressions, the doping profile is generally approximated as a box-like or **abrupt junction** as shown in Fig 4.1(b), i.e. having constant dopant concentrations in both the n-type and p-type regions. We shall first discuss the electric field and potential distribution in such an abrupt p-n junction and later extend the analysis to linearly graded junctions. Finally, the junction capacitances and the flow of current in a p-n junction under applied bias will be considered.

4.1 The p-n junction under thermal equilibrium

We have already seen in Chapter 1 that the Fermi level in a p-type semiconductor lies close to the valence band edge, while in an n-type semiconductor, it is close to the conduction band edge. Physically, this means that the p-region has a large concentration of holes (and few electrons) while n-region has a large concentration of electrons (and few holes). Now, when a p-region and an n-region are brought in intimate contact (i.e. a junction is formed), this large concentration gradient at the junction causes diffusion of carriers. While holes diffuse from the p-side to the n-side, electrons diffuse from the n-side to the p-side. This process results in some uncompensated donor ions (N_D^+) on the n-side and some uncompensated acceptor ions (N_A^-) on the p-side near the junction. This is

schematically shown in Fig.4.2(a). Consequently, a **negative space charge** builds up on the p-side and a **positive space charge** on the n-side. This creates a built-in electric field directed from positive charge to negative charge (i.e. from n-side to p-side) which gives rise to a drift current. The direction of this drift current will be opposite to that of the diffusion current for both electrons and holes as shown in Fig. 4.2(b). An equilibrium condition is reached where there is no nett transport of carriers as the diffusion component of current is exactly balanced by an equal and opposite drift component.

Also, since there can be no nett build-up of charge on either side of the junction as a function of time, the drift and diffusion components must cancel each other for **each** type of charge carrier, i.e. J_p and J_n must each be equal to zero. If that is not the case, then to satisfy the requirement of no net current, J_p must be equal to $-J_n$. This will signify that both electrons and holes are moving in the same direction, which is absurd. Therefore,

$$J_p = J_{pdr} + J_{pdiff} = 0 \ldots\ldots\ldots (4.1a)$$

and

$$J_n = J_{ndr} + J_{ndiff} = 0 \ldots\ldots\ldots (4.1b)$$

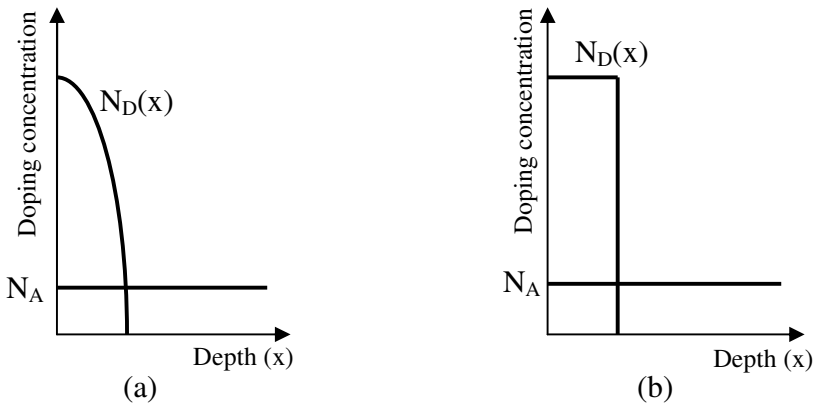


Fig. 4.1(a) Actual doping profile in a p-n junction and (b) the approximate abrupt doping profile

Also, we know that for any junction at thermal equilibrium, the Fermi level must be constant throughout. The energy band diagram for this abrupt p-n junction can therefore be drawn simply by aligning the Fermi level as shown in Fig. 4.2(c). Far from the junction, the electron and hole concentrations on both sides remain unchanged and hence the positions of E_V and E_C with respect to the Fermi level also remain the same as they were before the junction was formed. In the space charge region, however, the conduction and valence band edges bend, signifying the presence of an electric field. Since the energy band diagram reflects the electronic potential (as discussed in Section 1.2.4, the electronic potential is the negative of the electrostatic potential), the n-side is at a higher electrostatic potential than the p-side. This difference in E_C (or E_V) from the p-side to the n-side is given by qV_{bi}, where V_{bi} is the called the **contact potential** or the **built-in potential.**

We have seen in Fig. 1.9 that electrons at a higher potential energy can easily "fall down" to a region of lower potential energy in the conduction band. On the other hand, since hole energy in an energy band diagram increases downwards, holes can easily "climb up" a potential energy variation in the valence band. However, the situation in Fig. 4.2(c) is quite different. There is a large concentration of electrons in the conduction band on the n-side and holes in the valence band on the p-side. These electrons have to "climb up" while these holes have to "fall down" in order to cross the junction. This is possible only if the carriers have sufficiently large kinetic energy. The potential difference across the space charge layer therefore creates a barrier to the flow of carriers called the "**potential energy barrier**". This barrier is equal to qV_{bi} at thermal equilibrium. We shall see later that this barrier can be modified by application of bias.

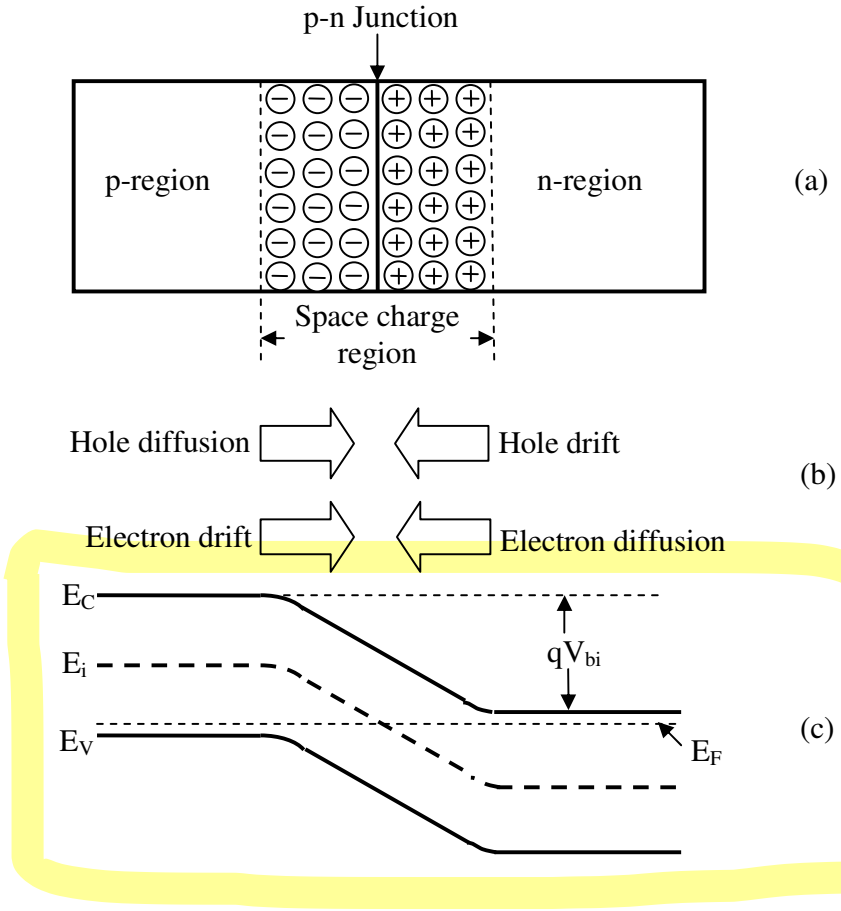


Fig. 4.2(a) A p-n junction showing the uncompensated acceptor and donor impurities in the space charge region; (b) the directions of flow of electrons and holes due to drift and diffusion and (c) the band diagram at thermal equilibrium

4.1.1 Calculation of built-in potential

The built-in potential can be calculated as the difference in the positions of the intrinsic levels in the n and p-sides of the junction in terms of the doping concentrations in the following manner. On the p-side, using eqn. (1.30), the position of the Fermi level is given by

$$E_{Fp} = E_{ip} - kT \ln \frac{N_A}{n_i} \quad \text{.....(4.2a)}$$

Similarly, on the n-side

$$E_{Fn} = E_{in} + kT \ln \frac{N_D}{n_i} \quad \text{.....(4.2b)}$$

Since the Fermi level is constant throughout, $E_{Fp} = E_{Fn}$. Therefore, from eqn. (4.2), we have

$$E_{ip} - E_{in} = kT \ln \frac{N_A N_D}{n_i^2} = qV_{bi} \quad \text{.....(4.3)}$$

Assuming complete ionization, we can write $p_{po} = N_A$, $n_{no} = N_D$, $n_{po} = n_i^2/p_{po}$ and $p_{no} = n_i^2/n_{no}$. So, the built-in potential can be expressed as

$$V_{bi} = V_T \ln \frac{p_{po}}{p_{no}} = V_T \ln \frac{n_{no}}{n_{po}} \quad \text{.....(4.4)}$$

where p_{po} and n_{no} are the majority carrier concentrations and p_{no} and n_{po} are the minority carrier concentrations at thermal equilibrium.

Example 4.1 Calculate the built-in potential at 300K for an abrupt silicon p-n junction with $N_A = 10^{18}/cm^3$ and $N_D = 10^{15}/cm^3$ on the p-side and n-side respectively. Also find the values of the built-in potential when the semiconductor is (a) Ge and (b) GaAs.

Solution: For silicon, the intrinsic carrier concentration n_i at 300K is $1.5 \times 10^{10}/cm^3$. Also kT/q at 300K is 0.026V. Using equation 4.3, and substituting the values of n_i, N_A and N_D, we have $V_{bi} = 0.026 \ln \frac{10^{18} \times 10^{15}}{(1.5 \times 10^{10})^2} = 0.76V$.

For Ge, n_i at 300K is $2.4 \times 10^{13}/cm^3$. Therefore $V_{bi} = 0.026 \ln \frac{10^{18} \times 10^{15}}{(2.4 \times 10^{13})^2} = 0.37V$.

For GaAs n_i at 300K is $1.79 \times 10^6/cm^3$. Therefore $V_{bi} = 0.026 \ln \frac{10^{18} \times 10^{15}}{(1.79 \times 10^6)^2} = 1.23V$

From the above example we see that for the same extrinsic doping concentrations on the p and n-sides of the junction, the built-in potential is larger for materials with larger bandgap and consequently smaller intrinsic carrier concentration. In this example $V_{bi}(GaAs) > V_{bi}(Si) > V_{bi}(Ge)$.

Q.4.1: Can the built-in potential be measured directly?

A built-in potential or contact potential is developed whenever two dissimilar materials are brought in contact. The two materials may be p-type and n-type regions of the same semiconductor, or a metal and a semiconductor. In all such cases, as the Fermi levels of the two materials in contact should align, the built-in potential (in V) is equal to the difference in the positions of the Fermi level (in eV) of the two materials when they were isolated, or in other words to the difference in the positions of the work functions. Now if we are to measure the contact potential, we have to connect a voltmeter, which requires that metal contacts be provided at both p and n ends of the junction. We now have three junctions, viz. metal-p-semiconductor, p-n and n-semiconductor-metal junctions, in series. The voltmeter will measure the sum of the three contact potentials. If $q\phi_m$, $q\phi_n$ and $q\phi_p$ are the work functions of the metal, n-semiconductor and p-semiconductor respectively, the voltmeter reading will be equal to $V = \frac{(q\phi_m - q\phi_p) + (q\phi_p - q\phi_n) + (q\phi_n - q\phi_m)}{q} = 0$. Thus we see that the built-in potential cannot be measured by simply connecting a voltmeter across the junction as the sum of the contact potentials between metal and semiconductor exactly compensate the built-in potential of the p-n junction.

4.1.2 Concept of Space Charge Layer

It has already been pointed out earlier in Section 4.1 that due to the concentration gradient existing at the junction, a space charge layer (also called transition layer) will be created with uncompensated donor ions on the n-side and acceptor ions on the p-side. Within this space charge region, electrons and holes are in transit from one side to the other. However, it is important to note that there are much fewer charge carriers than dopants in most of this transition region. So, we can consider that within the space charge region the charge is only due to the immobile acceptor and donor ions. In other words, this region is depleted of mobile carriers and is also referred to as **depletion layer**. In this depletion region, as we move from the p-side to n-side, the difference between the intrinsic level and Fermi level (E_i-E_F) at first reduces, becomes zero and then becomes negative as can be seen from Fig. 4.2(c). This actually means that the hole concentration across this region decreases from p_p on the p-side to p_n on the n-side. Similarly, the concentration of electrons decreases from n_n on the n-side to n_p on the p-side. Fig. 4.3 shows the variation of the carrier concentrations in the space charge region.

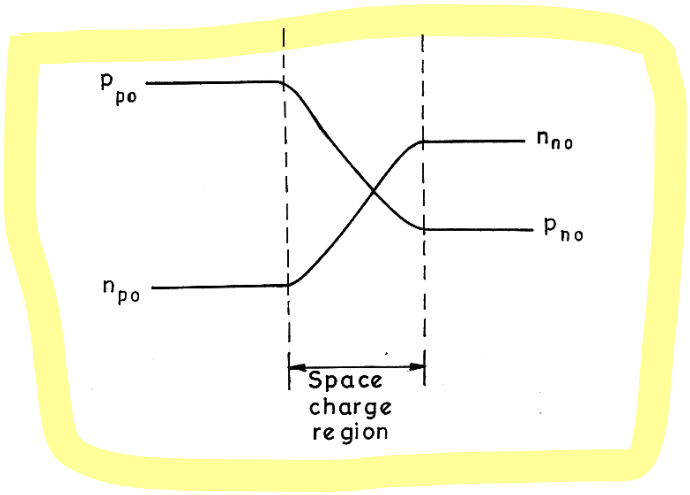


Fig. 4.3 Variation of electron and hole concentrations in the space charge region

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Example 4.2: Consider a silicon p-n junction where the doping concentration on the p-side is $10^{18}/cm^3$. Calculate the space charge concentration at a point 'x' inside the space charge region where $\psi(x) = 0.1V$. Assume that this point 'x' lies on the p-side.

Solution: Assuming complete ionization, the equilibrium hole concentration on the p-side $p_{po} = N_A = 10^{18}/cm^3$ and the electron concentration $n_{po} = n_i^2/N_A = 2.25x10^2/cm^3$ at 300K.

From eqn.(1.28), we can write $p_{po} = n_i exp\left(\frac{E_{ip} - E_F}{kT}\right)$ and $p(x) = n_i exp\left(\frac{E_i(x) - E_F}{kT}\right)$.

Therefore, from these equations, we can write

$$p(x) = p_{po} exp\left(\frac{E_i(x) - E_{ip}}{kT}\right) = p_{po} exp\left(\frac{-q\psi_S}{kT}\right)$$

Now, at the point x where $\psi(x) = 0.1V$, the hole concentration p(x) is given by

$$p(x) = 10^{18} exp\left(\frac{-0.1}{0.026}\right) = 2.13x10^{16}/cm^3$$

Following a similar procedure for electrons using eqn.(1.27), it can be shown that

$$p(x) = 10^{18} exp\left(\frac{-0.1}{0.026}\right) = 2.13x10^{16}/cm^3$$

$$n(x) = n_{po} exp\left(\frac{q\psi_S}{kT}\right) = 2.25x10^2 exp\left(\frac{0.1}{0.026}\right) = 1.05x10^4/cm^3$$

Thus the total space charge density at the point x is $Q(x) = -q[N_A - p(x) + n(x)]$

$= -q\,[10^{18} - 2.13x10^{16} + 1.05x10^4\,] = -q\,9.98x10^{17} \approx -q10^{18} = -qN_A$

Example 4.2 serves as a validation of the **depletion approximation**, according to which the number of mobile charges inside the space charge layer is negligible. It is evident that

the hole and electron concentrations are much smaller compared to the ionized dopant concentration and hence can be ignored. Thus, neglecting mobile carriers within the space charge, the charge density on the p-side is $\rho_p = -qN_A$ and on the n-side it is $\rho_n = qN_D$.

It must, however, be noted that at the edges of the space charge layer, the mobile carrier concentration approaches the concentration of the dopants, and hence depletion approximation may not be valid. (For example, consider $\psi(x) = 0.026V$ in Example 4.2). However, the thickness of this region is very small compared to the overall space charge layer thickness. Hence, in all further analysis in this book, for the sake of simplicity, we shall assume that the depletion approximation is valid throughout the space charge region. *In other words, we assume that the space charge region has sharp boundaries, where the charge concentration changes abruptly from* $\rho_p = -qN_A$ *to zero on the p-side and* $\rho_n = qN_D$ *to zero on the n-side.*

4.1.3 Distribution of Electric field and Potential within the space-charge layer for abrupt junctions at zero bias

As the electric field lines must begin and end on charges of opposite sign, the total number of uncompensated acceptor ions on the p-side must be equal to that of the uncompensated donors on the n-side of the depletion layer. This causes the space-charge layer to extend unequally into the p and n-regions depending upon the relative doping concentrations of both sides. For example, if the doping concentration in the p-side (N_A) is greater than the doping concentration in the n-side (N_D), then the space charge region will extend further into the n region than into p to uncover equal amount of charge. Thus the total uncompensated charge on each side of the junction (Q_D) is given by

$$Q_D = AqN_A x_p = AqN_D x_n \quad \ldots\ldots\ldots\ldots (4.5)$$

where A is the cross-sectional area of the junction, x_p is the penetration of depletion region into p-region, x_n is the penetration of depletion region into n-region and q is the electronic charge. In this book, we shall consider only one-dimensional analysis for simplicity, thereby assuming that there is no variation in the device properties along the cross-section. A p-n junction, where the depletion region extends from $x = -x_p$ to $x = x_n$, is shown schematically in Fig. 4.4(a). The corresponding charge (ρ) distribution is shown in Fig. 4.4(b). Since the number of acceptor impurities in the space charge region on the p-side is equal to the donor impurities on the n-side, the area under the charge distribution curve on the p-side (given by $N_A x_p$) must be equal to that on the n-side (given by $N_D x_n$). From eqn. (4.5) as well as Fig. 4.4(b), a relationship between the relative doping concentration and the penetration of depletion region can be obtained as

$$\frac{x_n}{x_p} = \frac{N_A}{N_D} \quad \ldots\ldots\ldots (4.6)$$

In other words, if the doping concentration on the p-side of the junction is 1000 times the doping concentration on the n-side, the width of the depletion region on the n-side will be 1000 times that on the p-side. In such cases, for all practical purposes, the depletion region exists only on one side of the junction and the penetration on the heavily doped side can be neglected. Such a junction is called a **one-sided abrupt junction** and will be used in our analysis later in this chapter.

Let us now again refer to Fig. 4.4 and obtain an expression for the electric field distribution inside the depletion region. To relate the charge distribution with the electrostatic potential (ψ), we start with **Poisson's equation**, which can be derived **Gauss**

Law. According to Gauss Law, the total normal electric flux coming out of a closed surface is equal to the charge enclosed by it. This leads to the relation

$$\nabla.(\varepsilon_s \varepsilon) = \rho \ldots\ldots..(4.7a)$$

which is one of Maxwell's equations. In eqn.(4.7a), ρ is the charge density in the space charge layer, ε_s is the permittivity of the semiconductor and ε is the electric field given by $\varepsilon = -\nabla\psi$. Considering the existence of electric field only in the x-direction, the one-dimensional Poisson's equation can now be derived as

$$\frac{d^2\psi}{dx^2} = -\frac{\rho}{\varepsilon_s} \ldots\ldots\ldots(4.7b)$$

Neglecting the presence of mobile carriers in the space charge region, eqn.(4.7b) reduces to

$$\frac{d\varepsilon}{dx} = \frac{q(N_D - N_A)}{\varepsilon_s} \ldots\ldots.(4.8)$$

Referring to Fig 4.4(b), we see that

$$\frac{d\varepsilon}{dx} = \frac{qN_D}{\varepsilon_s} \text{ for } 0 < x < x_n \ldots\ldots.(4.9a)$$

$$\frac{d\varepsilon}{dx} = -\frac{qN_A}{\varepsilon_s} \text{ for } -x_p < x < 0 \ldots\ldots(4.9b)$$

Thus, a plot of $\varepsilon(x)$ as a function of x within the depletion region has two slopes as seen from Fig.4.4(c). The slope is positive on the n-side and negative on the p-side. Also, the magnitude of the field is maximum at x = 0. Physically, the reason for this is that **all** the field lines originating from the positive charges and terminating on the negative charges must pass through x = 0. This is not true for any other point in the space charge region. The negative value of the electric field in Fig. 4.4(c) refers to its direction, which is from

the n-side to the p-side, i.e. in the negative x-direction. Also, at the edge of the space charge region, electric field is assumed to fall to zero as there are no charges outside the depletion region.

Integrating eqn. (4.9a) with respect to x, we get an expression for the electric field on the n-side as

$$\varepsilon(x) = \int \frac{qN_D}{\varepsilon_s} dx = \frac{qN_D x}{\varepsilon_s} + C \quad \text{.....(4.10)}$$

where C is the integration constant. C can be evaluated by using the condition $\varepsilon(x_n) = 0$. Therefore, we have $C = -\frac{qN_D x_n}{\varepsilon_s}$. Substituting the value of C in eqn. (4.10), we have

$$\varepsilon_n(x) = \frac{qN_D(x - x_n)}{\varepsilon_s} = \varepsilon_m\left(1 - \frac{x}{x_n}\right) \quad \text{for } 0<x<x_n \text{ ...(4.11)}$$

where $\varepsilon_m = -\frac{qN_D x_n}{\varepsilon_s}$ is the maximum electric field at x = 0.

Following a similar analysis as above on the p-side, we have

$$\varepsilon_p(x) = -\frac{qN_A(x + x_p)}{\varepsilon_s} = \varepsilon_m\left(1 + \frac{x}{x_p}\right) \quad \text{for } -x_p< x < 0\text{.....(4.12)}$$

where $\varepsilon_m = -\frac{qN_A x_p}{\varepsilon_s}$ is the maximum electric field at x = 0. Using eqn.(4.5), we can show that the expressions for ε_m obtained from the n and p sides are indeed equivalent, or

$$\varepsilon_m = -\frac{qN_D x_n}{\varepsilon_s} = -\frac{qN_A x_p}{\varepsilon_s} \quad \text{.....(4.13)}$$

It may be noted that ε_m is proportional to the area under the charge concentration profiles in Fig. 4.4(b). Now, integrating eqns. (4.11) and (4.12) over the entire space charge region, we obtain the built-in potential as

$$V_{bi} = -\int_{-x_p}^{x_n} \varepsilon dx = -\int_{-x_p}^{0} \varepsilon_p dx - \int_{0}^{x_n} \varepsilon_n dx = \frac{q(N_A x_p^2 + N_D x_n^2)}{2\varepsilon_s} = \frac{|\varepsilon_m| W}{2} \quad .....(4.14)$$

where W (= x_p+x_n) is the total width of the space charge layer. It can be easily verified from Fig.4.4(c) that the built-in potential, i.e. the potential dropped across the depletion layer is given by the area of the triangle covered by the ε(x) vs. x plot. This area can also be viewed to be the sum of two triangles having the same altitude (=ε_m) but different base lengths (x_n and x_p). The areas of these two triangles signify the potential dropped across the n-side of the depletion layer (V_n) and that dropped across the p-side of the depletion layer (V_p). The values of V_n and V_p calculated geometrically from the areas of the triangles agree perfectly with eqn.(4.14), which has been derived analytically. In fact, obtaining the electric field distribution and subsequently the potential variation from the doping concentration profile geometrically is often more convenient than analytical methods.

Now, V_n and V_p are related by the following expression

$$\frac{V_n}{V_p} = \frac{x_n}{x_p} = \frac{N_A}{N_D} \quad .....(4.15)$$

From eqn. (4.15), we can write $x_p = \frac{N_D W}{N_A + N_D}$ and $x_n = \frac{N_A W}{N_A + N_D}$. Substituting these values of x_p and x_n in eqn.(4.14), we get

$$W = \sqrt{\frac{2\varepsilon_s V_{bi}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}{q}} = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_{eff}}} \quad ... (4.16)$$

where N_{eff} can be considered to be an effective doping concentration given by

$$\frac{1}{N_{eff}} = \frac{1}{N_A} + \frac{1}{N_D} \quad \text{.....(4.17)}$$

For a one-sided abrupt junction, if $N_A >> N_D$, $N_{eff} \approx N_D$, and the above equation can be approximated as

$$W = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_D}} \quad \text{.....(4.18)}$$

This also implies that virtually the entire built-in potential is dropped across the lightly doped side.

Example 4.3: Calculate the maximum electric field and the width of the depletion region at zero bias for an abrupt silicon p-n junction with $N_A = 10^{19}/cm^3$ and $N_D = 10^{15}/cm^3$ at room temperature. (Use $n_i = 1.5 \times 10^{10}/cm^3$, $\varepsilon_r = 11.9$ for Si, $\varepsilon_o = 8.85 \times 10^{14}$F/cm)

Solution: Using eqn. (4.3), we have $V_{bi} = 0.026 \ln \frac{10^{19} \times 10^{15}}{(1.5 \times 10^{10})^2} = 0.817\text{V}$

In this problem, since N_A/N_D is 10^4, we consider the junction to be a one-sided abrupt junction. Now, from eqn. (4.18),

$$W = \sqrt{\frac{2 \times 11.9 \times 8.85 \times 10^{-14} \times 0.817}{1.6 \times 10^{-19} \times 10^{15}}} = 1.037 \times 10^{-4}\text{cm} = 1.037\mu\text{m}.$$

From eqn.(4.14), $\varepsilon_m = \frac{2V_{bi}}{W} = \frac{2 \times 0.817}{1.037 \times 10^{-4}} = 1.58 \times 10^4$ V/cm

It is left as an exercise for the student to verify that the depletion layer width will be very nearly the same, if one does not consider a one-sided junction and uses eqn. (4.16).

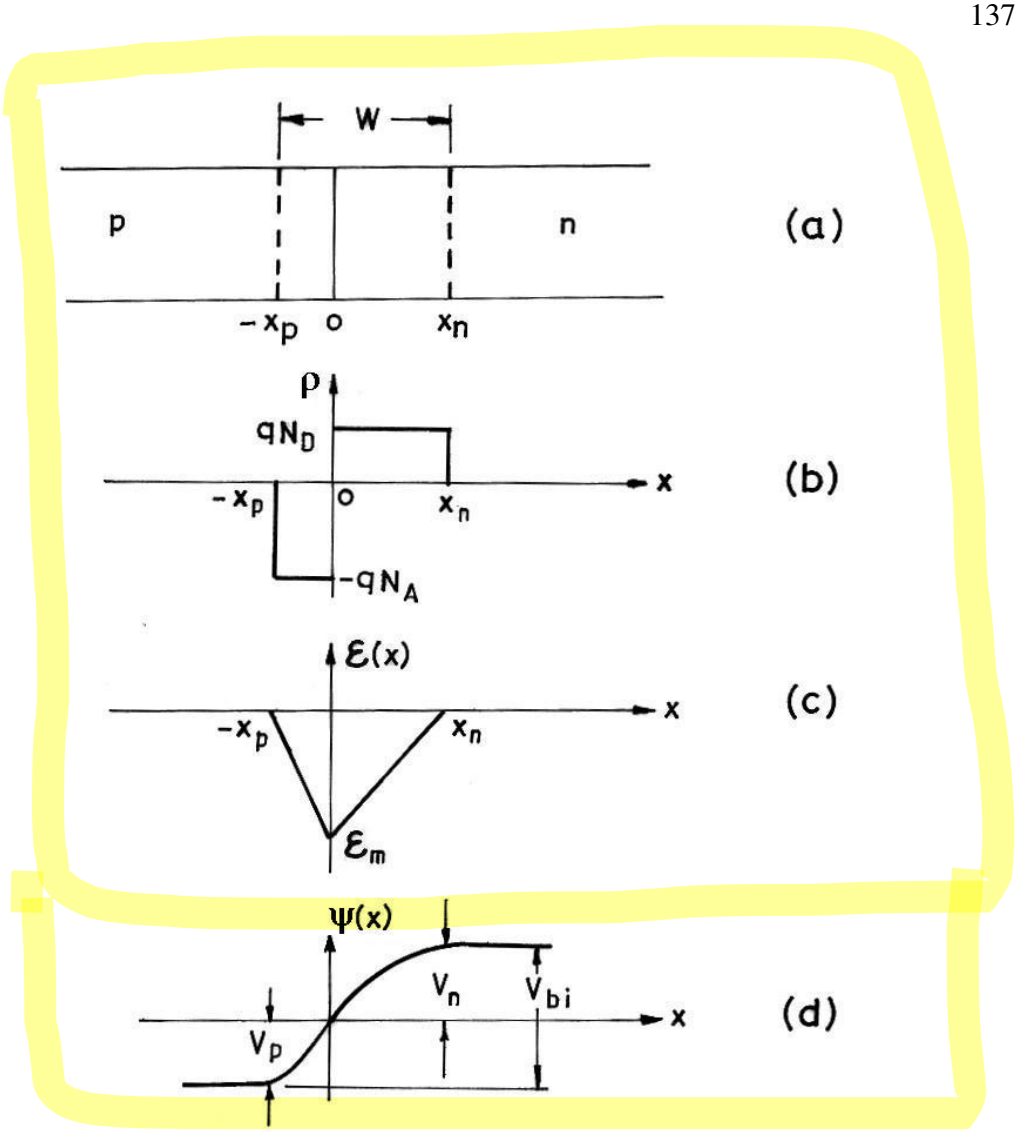


Fig. 4.4 A p-n junction showing (a) the space charge region from x = $-x_p$ to x = x_n, (b) the charge distribution, (c) the electric field profile and (d) the potential variation

4.1.4 Distribution of Electric field and Potential within the space-charge layer for linearly graded junctions at zero bias

For many practical situations, the approximation of abrupt doping profile may not be valid. Let us now consider the case of a **linearly graded junction** where the doping concentration is a linear function of x as shown in Fig.4.5(a). For such a situation

$$N_D - N_A = ax \quad \text{.....(4.19)}$$

where a is a constant of proportionality. Then the solution of Poisson's equation yields an expression for the electric field as

$$\varepsilon(x) = \frac{qax^2}{2\varepsilon_s} + C \ldots\ldots\ldots (4.20)$$

where C is a constant of integration to be evaluated from the boundary condition that at the edges of the depletion layer ($x = \pm W/2$) the electric field falls to zero. It may be noted that since the doping concentration is symmetric on either side of the junction, the depletion width will extend equally on both sides of the junction, i.e. $x_n = x_p = W/2$. Therefore

$$\varepsilon(x) = \frac{qa}{2\varepsilon_s}\left[x^2 - \frac{W^2}{4}\right] \ldots\ldots\ldots (4.21)$$

The plot of $\varepsilon(x)$ as a function of x is shown in Fig.4.5(b). From eqn. (4.22), the value of ε_m can be evaluated as

$$\varepsilon_m = \varepsilon\big|_{x=0} = -\frac{qaW^2}{8\varepsilon_s} \ldots\ldots\ldots (4.22)$$

Integrating eqn.(4.21), we obtain the expression for the built-in potential as

$$V_{bi} = \int_{-x_p}^{x_n} \varepsilon(x)dx = \frac{qaW^3}{12\varepsilon_s} \ldots\ldots\ldots (4.23)$$

It can be shown that the built-in potential is equal to the area under the $\varepsilon(x)$ curve in Fig. 4.5(b). Also, we know that the built-in potential is actually the difference in the Fermi level positions with respect to the intrinsic levels at the two edges of the depletion layer, i.e.

$$qV_{bi} = \left(E_F - E_i(x)\right)\big|_{x=W/2} - \left(E_F - E_i(x)\right)\big|_{x=-W/2} \ldots\ldots\ldots (4.24)$$

Writing (E_F-E_i(x)) in terms of the doping concentration as in eqns. (1.29) and (1.30) at the edges of the depletion layer, we have

$$V_{bi} = V_T \ln\left(\frac{(aW/2)(aW/2)}{n_i^2}\right) = 2V_T \ln\left(\frac{aW}{2n_i}\right) \ldots\ldots\ldots (4.25)$$

Solving eqns. (4.23) and (4.25) simultaneously, it is possible to obtain numerical values for the depletion layer width and built-in potential for a particular case of linearly graded junction. However, this cannot be done analytically and one has to resort to numerical techniques.

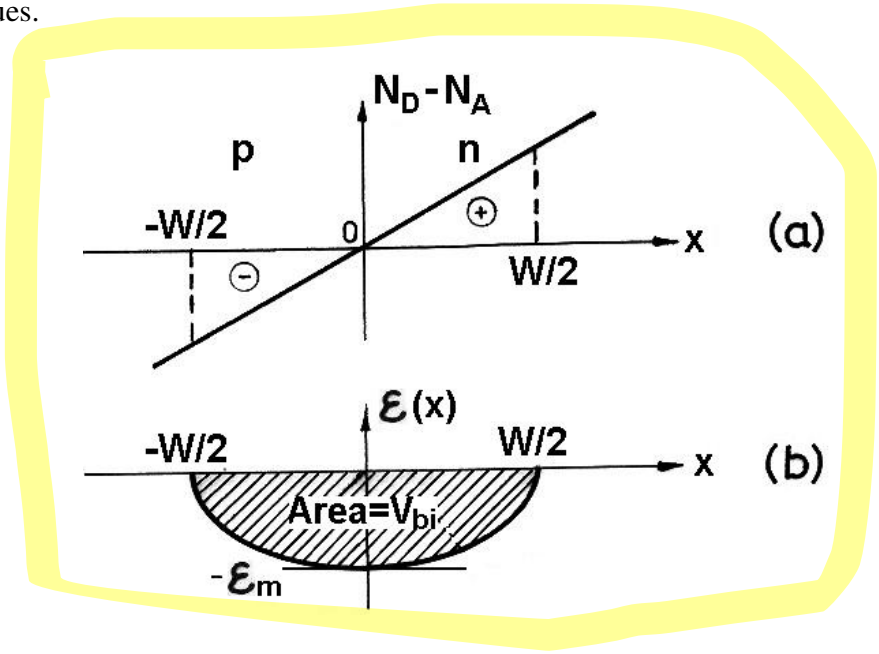


Fig. 4.5 (a) The doping concentration profile and (b) the electric field variation in a linearly graded p-n junction

4.2 The p-n junction under applied bias

When an external voltage is applied to the p-n junction, the electron and hole concentrations change from their equilibrium values. The diode current is intimately related to these changes. Since the space charge region is devoid of mobile carriers, its resistance is much greater than that of the neutral p and n regions. Therefore practically all the applied voltage is dropped across this space charge layer and is superimposed on

the built-in potential. In other words, the potential difference across the depletion layer is changed from its equilibrium value of V_{bi} by the amount of applied bias. When the p-n junction is **forward biased** by V_f, i.e. the positive terminal of the battery is connected to the p-side and the negative terminal to the n-side, the potential difference across the space charge region is reduced to $(V_{bi} - V_f)$ as shown in Fig. 4.6(a). On the other hand, when the p-n junction is **reverse biased** by V_r, i.e. the positive terminal of the battery is connected to the n-side and the negative terminal to the p-side, the potential difference is increased to $(V_{bi} + V_r)$ as shown in Fig. 4.6(b). Fig. 4.6 also shows the variation in Quasi Fermi levels, which will be discussed in Section 4.3.2.

When a forward voltage is applied, the direction of the applied field is opposite to that of the built-in field. Consequently, the effective field across the junction reduces with forward bias. Conversely, when a reverse bias is applied, the effective field is increased. This change in the electric field also affects the depletion layer width. For forward voltages, the electric flux lines reduce, leading to fewer uncompensated charges and smaller W. By the same logic, application of reverse voltage results in increase in electric flux lines and therefore a larger W. Since the voltage drop across the junction is now $(V_{bi} - V)$, for an abrupt junction, the new values of W and E_m can be obtained by replacing V_{bi} with $(V_{bi} - V)$ in eqns. (4.14) and (4.16), i.e.

$$|\varepsilon_m| = \frac{2(V_{bi} - V)}{W} \quad \text{.....(4.26a)}$$

$$W = \sqrt{\frac{2\varepsilon_s (V_{bi} - V)}{qN_{eff}}} \quad \text{.....(4.26b)}$$

where V is the applied voltage (equal to V_f for forward bias and $-V_r$ for reverse bias).

Example 4.4: For the p-n junction of Example 4.3, obtain the maximum electric field and depletion layer width when (a) a forward voltage of 0.3V and (b) a reverse voltage of 3V is applied. Also sketch the electric field distribution for these two cases along with that in thermal equilibrium as obtained in Ex.4.3.

Solution: In Example 4.3, we have calculated $V_{bi} = 0.817V$

Case (a) When $V_f = 0.3V$, $V_{bi}-V_f = 0.817-0.3 = 0.517V$

Now, from eqn. (4.26b), $W = \sqrt{\frac{2x11.9x8.85x10^{-14}\,x0.517}{1.6x10^{-19}\,x10^{15}}} = 8.25x10^{-5}\,cm = 0.825\mu m.$

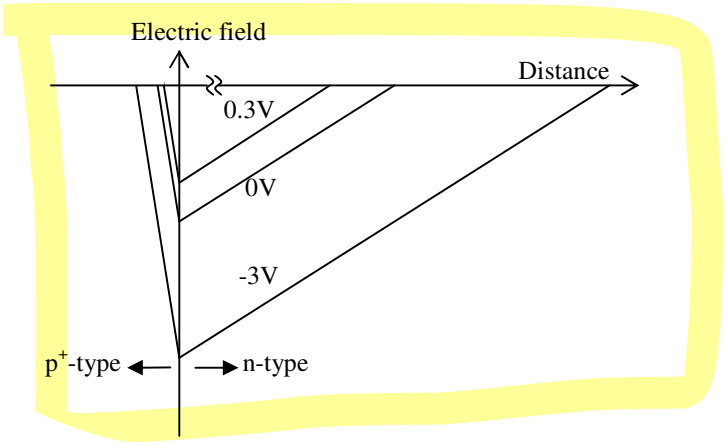
From eqn.(4.26a), $|\varepsilon_m| = \frac{2(V_{bi}-V)}{W} = \frac{2x0.517}{8.25x10^{-5}} = 1.25x10^{4}\,V/cm$

Case (b) When $V_r = 3V$, $V_{bi}+V_r = 0.817+3.0 = 3.817V$

Now, from eqn. (4.26b), $W = \sqrt{\frac{2x11.9x8.85x10^{-14}\,x3.817}{1.6x10^{-19}\,x10^{15}}} = 2.24x10^{-4}\,cm = 2.24\mu m.$

From eqn.(4.26a), $|\varepsilon_m| = \frac{2(V_{bi}-V)}{W} = \frac{2x3.817}{2.24x10^{-4}} = 3.4x10^{4}\,V/cm$

The sketch of the electric field distributions for two cases above along with the thermal equilibrium case is shown below:



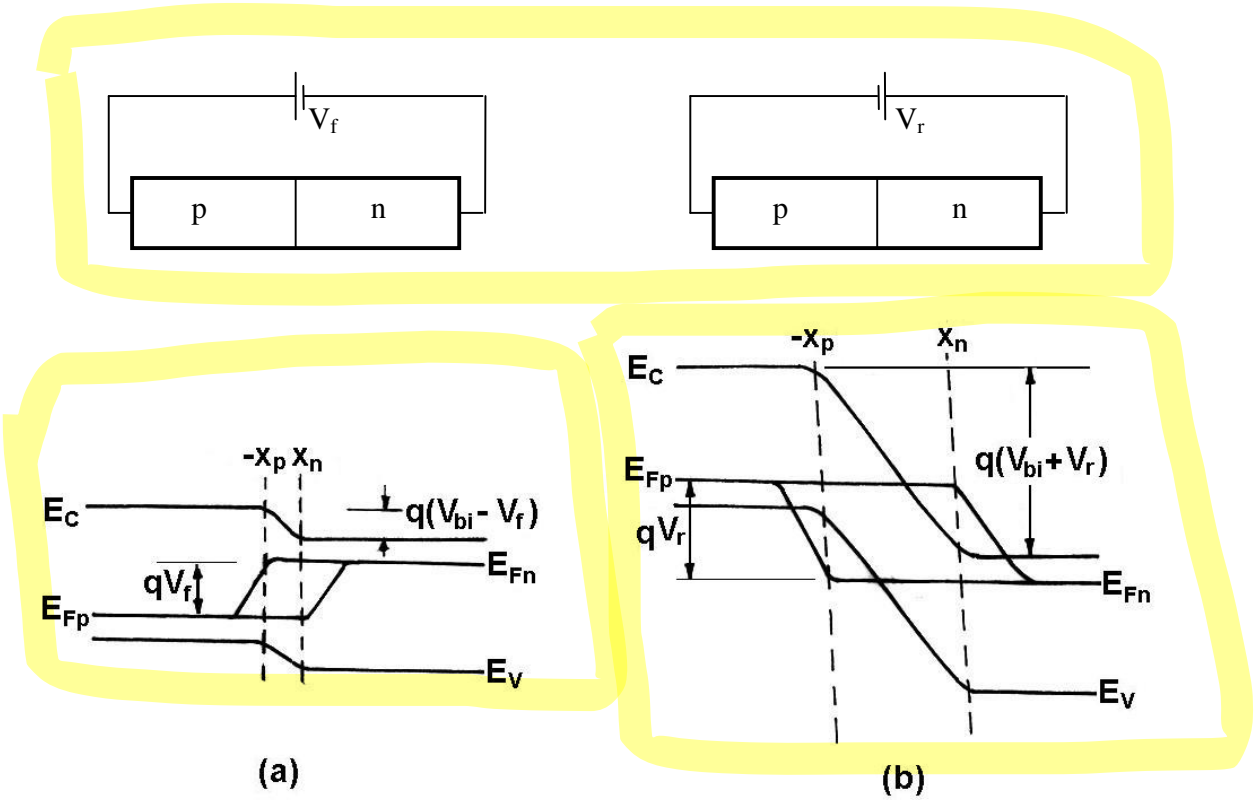


Fig.4.6 Biasing and energy band diagram for a p-n junction under (a) forward bias and (b) reverse bias

4.2.1. Depletion Layer capacitance in an abrupt p-n junction

From the above discussion regarding space-charge layer, we can visualize it as a dipole layer of positive donor and negative acceptor ions. As the bias voltage is changed, the width of the depletion layer also changes, as majority carriers flow in or out of this layer. This is similar to the charging and discharging of a capacitor. The **depletion capacitance**, also called the **junction capacitance (C_J)** is thus expressed as

$$C_J = \frac{dQ_D}{dV} \quad \text{.....(4.27)}$$

Using eqns. (4.5), (4.6) and (4.17), we can write

$$Q_D = AqN_{eff}W \quad \text{.....(4.28)}$$

Now from eqns. (4.27), (4.28) and (4.26b), we have

$$C_J = \frac{dQ_D}{dV} = \frac{dQ_D}{dW}\frac{dW}{dV} = AqN_{eff}\sqrt{\frac{\varepsilon_s}{2qN_{eff}(V_{bi}-V)}} = A\varepsilon_s\sqrt{\frac{qN_{eff}}{2\varepsilon_s(V_{bi}-V)}} \quad .....(4.29)$$

Substituting eqn.(4.26) in eqn.(4.29), we now have

$$C_J = \frac{A\varepsilon_s}{W} \quad .....(4.30)$$

Eqn.(4.30) is the well known expression for a parallel plate capacitor where the two plates are separated by a distance W. In other words, the expression for the depletion capacitance is identical to that of a parallel plate capacitor. However, unlike a parallel plate capacitor, here W varies nonlinearly with voltage and consequently, C_D also varies with a change in the applied bias. This change in C_D can be measured to extract the doping concentration of the substrate as well as the built-in potential especially in case of a one-sided abrupt p-n junction. For such a junction (p^+n), $N_{eff} \approx N_D$. Hence from eqn. (4.29), we can write

$$C_J = A\sqrt{\frac{q\varepsilon_s N_D}{2(V_{bi}-V)}} = C_{JO}\left(1-\frac{V}{V_{bi}}\right)^{-1/2} \quad .....(4.31)$$

where $C_J = C_{JO}$ when $V = 0$. From eqn.(4.31), we can write

$$\frac{1}{C_J^2} = \frac{2}{A^2 q\varepsilon_s N_D}(V_{bi}-V) \quad .....(4.32)$$

So we find that if we change the bias voltage across the p-n junction diode and measure the capacitance (C_J) at each bias point and finally plot ($1/C_J^2$) versus the applied voltage, we will get a straight line with a slope equal to $-\frac{2}{A^2 q\varepsilon_s N_D}$ as shown in Fig.4.7. Now, since q and ε_s are well known constants, if A is known we can find out the value of N_D from the slope of the straight line. It must be pointed out that the measurements must be

made with the diode in the reverse biased condition (i.e. V < 0), since in the forward biased condition the stored charge capacitance, which we shall discuss in Section 4.4.3, dominates and the measured capacitance is not equal to C_J. From eqn.(4.32), we find that when $V = V_{bi}$, $(1/C_J^2) = 0$. Therefore, extrapolating the $(1/C_J^2)$ versus V curve for positive V values, we get the value of V_{bi} from the intercept of the curve with the V-axis.

Thus we see that the reverse bias C-V measurement is an important tool for obtaining the doping concentration and built-in potential of a p-n junction.

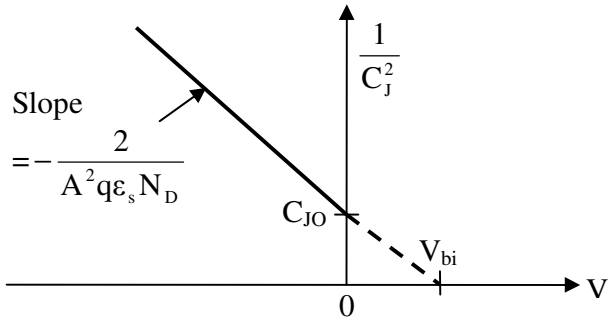


Fig. 4.7 $(1/C_J^2)$ versus V plots for reverse biased p-n junction

4.2.2. Depletion Layer capacitance in junctions with arbitrary doping profiles

Let us consider a p-n junction with an arbitrary doping profile as shown in Fig. 4.8(a). The electric field distribution for this profile at an applied bias V can be obtained by integrating the doping profile function, as discussed in Section 4.1.3, and is shown in Fig. 4.8(b). Now for an incremental increase in the applied reverse bias (-dV), there will be a corresponding change in the electric field (dε). If the resulting change in W is very small, i.e. dW<<W, we can see from the figure that $dV \approx W d\varepsilon$, where dV is given by the area of the shaded region in Fig. 4.8(b). From Gauss Law we know that the total electric flux coming out of a closed surface equals the charge enclosed by the surface. Therefore

$A\varepsilon_s d\varepsilon = dQ_D$, where $A\varepsilon_s d\varepsilon$ is the incremental total flux over the area A corresponding to the incremental depletion charge dQ_D. Now we have

$$C_J = \frac{dQ_D}{dV} = \frac{dQ_D}{Wd\varepsilon} = \frac{dQ_D}{W\left(\frac{dQ_D}{A\varepsilon_s}\right)} = \frac{A\varepsilon_s}{W} \quad \text{.....(4.33)}$$

Thus we see that eqn. (4.30) is also valid for any arbitrary doping profiles. However, for different doping distributions, the manner in which the depletion layer width changes with applied reverse bias will be different and thus the voltage sensitivity of the depletion layer capacitance will also be different. This will be discussed further in Chapter 5.

For an incremental change in Q_D caused by an incremental change in the applied reverse bias (-dV), we can write in general, for a one-sided junction

$$dQ_D = AqN(W)dW \quad \text{..... (4.34)}$$

where N(W) is the doping concentration at x =W, i.e. at the edge of the depletion region. Now, from the above discussion and using eqn.(4.34), we can write

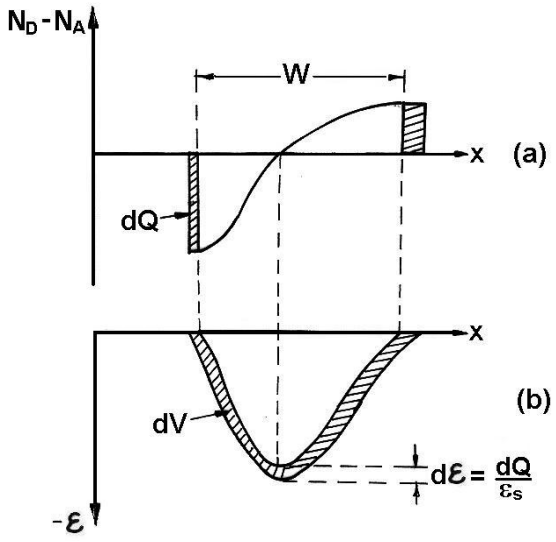


Fig. 4.8 (a) Doping profile in an arbitrarily doped p-n junction and
(b) the corresponding electric field variation

$$-dV = Wd\varepsilon = W\left(\frac{dQ_D}{A\varepsilon_s}\right) = \frac{qN(W)WdW}{\varepsilon_s} \quad \ldots\ldots(4.35)$$

$$WdW = \tfrac{1}{2}\left(dW^2\right) = \tfrac{1}{2}d\left(\frac{A^2\varepsilon_s^2}{C_J^2}\right) = \frac{A^2\varepsilon_s^2}{2}d\left(\frac{1}{C_J^2}\right) \quad \ldots\ldots.(4.36)$$

Now substituting eqn.(4.36) in eqn.(4.35), we have

$$-dV = \frac{qN(W)A^2\varepsilon_s^2}{2\varepsilon_s}d\left(\frac{1}{C_J^2}\right) \quad \text{or} \quad \frac{d\left(\frac{1}{C_J^2}\right)}{dV} = -\frac{2}{qN(W)A^2\varepsilon_s} \quad \ldots\ldots.(4.37)$$

Eqn.(4.37) shows that the capacitance-voltage (C-V) characteristics of p-n junctions can also be used to obtain the doping profile for non-uniformly doped regions. The initial process is the same as for uniformly doped junctions, i.e. after C-V measurements, the $(1/C_J^2)$ versus V curve is plotted. From eqn.(4.37), we see that from the slope of this curve, we get the doping concentration at the edge of the depletion region N(W) corresponding to a particular C_J. Using eqn.(4.33), we can obtain W for each value of C_J, and therefore we get the N(W) versus W variation. It can be easily verified that for uniform doping concentration, i.e. N(W) = constant, eqn. (4.37) reduces to eqn. (4.32).

4.3 Static current-voltage characteristics of p-n junctions

At zero bias, there is no nett transfer of carriers across the junction. With the application of forward bias, the potential energy barrier is lowered as shown in Fig. 4.6(a). A large number of electrons from the n-side and holes from the p-side are now able to diffuse across the junction as they have sufficient energy to overcome the reduced barrier. Thus the diffusion current in a forward biased junction can be quite large. The situation is different in a reverse biased junction, where an increase in the potential energy barrier reduces any possibility of diffusion of electrons from the n-side to the p-

side and holes from the p-side to the n-side. The current is now due to electrons from the p-side and holes from the n-side crossing the junction. Since these are minority carriers, and are therefore few in number, the reverse current is extremely small.

In this Section, we shall derive analytical expressions for the steady state current-voltage characteristics of p-n junction diodes in terms of the various device parameters.

4.3.1 Current-voltage relationship in an infinitely long diode

After the qualitative discussion regarding the p-n junction under bias presented in the previous section, let us now derive analytical expression for the current-voltage relation in an infinitely long diode. In doing so, we shall make the following assumptions:

1. The current flow across the junction is one dimensional, i.e. only in the x-direction.
2. All the applied voltage is assumed to drop across the space charge region. As the electric field in the neutral p and n regions is negligibly small, the minority carrier drift current can be neglected as shown in Section 3.7.
3. There is no net generation or recombination in the space charge layer and the current in this region remains constant throughout.
4. Low level injection condition prevails. This is valid for low currents.
5. Drift and diffusion currents almost balance each other in the depletion region even when a bias is applied.

While the other assumptions are easy to justify, the last one needs some explanation. Actually, the drift current is equal to the diffusion current only under zero bias (thermal equilibrium) condition. The hole diffusion current is given by $J_{pdiff} = qD_p \frac{dp}{dx}$. Assuming the depletion width to be 1μm, and the hole concentration difference to be about $10^{18}/cm^3$ across the depletion region, $J_{pdiff} \approx 10^4 A/cm^2$, which is a

large current density. This is balanced by an equally large hole drift current. Under bias conditions, the hole current flowing across the junction is the difference between the hole drift current and the hole diffusion current. Since the difference between the drift and diffusion components of the current itself is much smaller than the individual components under low injection levels, we can write even under bias

$$qp\mu_p \varepsilon \approx qD_p \frac{dp}{dx} \quad .....(4.38a)$$

Similarly, for electrons, we can write

$$qn\mu_n \varepsilon \approx qD_n \frac{dn}{dx} \quad .....(4.38b)$$

Since $\varepsilon = -\frac{d\psi}{dx}$, from eqn.(4.38a) and using Einstein relation, we have

$$\frac{d\psi}{dx} = -\frac{V_T}{p}\frac{dp}{dx} \quad .....(4.39)$$

Integrating between the edges of the depletion region, we can write

$$\int_0^{V_{bi}-V} d\psi = -V_T \int_{p_{pe}}^{p_{ne}} \frac{dp}{p} \quad .....(4.40)$$

where p_{pe} is the hole concentration at the edge of depletion region on the p-side and p_{ne} is the hole concentration at the edge of depletion region on the n-side as shown in Fig.4.9.

From eqn. (4.40), we have

$$p_{ne} = p_{pe} \exp\left(-\frac{V_{bi} - V}{V_T}\right) .....(4.41)$$

Substituting for V_{bi} from eqn. (4.4) into eqn. (4.41), we get

$$p_{ne} = \frac{p_{pe} p_{no}}{p_{po}} \exp\left(\frac{V}{V_T}\right) .....(4.42)$$

According to assumption 4, low-level injection condition prevails. That means the majority carrier concentration does not change much from its equilibrium value. Hence, $p_{pe} \approx p_{po}$. Eqn. (4.42) can thus be rewritten as

$$p_{ne} = p_{no} \exp\left(\frac{V}{V_T}\right) \ldots\ldots. \text{(4.43a)}$$

By analogy, for electrons in the p-type semiconductor, we have

$$n_{pe} = n_{po} \exp\left(\frac{V}{V_T}\right) \ldots\ldots\ldots \text{(4.43b)}$$

We now take a critical look at Fig. 4.9. It can be seen that when a forward bias is applied, the minority carrier concentrations at the edges of the depletion region (p_{ne} and n_{pe}) increase exponentially by a factor $\exp\left(\frac{V_f}{V_T}\right)$ from their equilibrium values. In other words, there is **minority carrier injection** due to forward bias. On the other hand, when a reverse bias is applied, p_{ne} and n_{pe} decrease exponentially by a factor $\exp\left(-\frac{V_r}{V_T}\right)$. This is because the minority carriers, which are close to the space charge region, diffuse to the space charge region and are then swept across by the strong electric field to the other side of the junction, resulting in **minority carrier extraction**. It may be mentioned here that along with the change in the minority carrier concentration, there is a corresponding change in the majority carrier concentration in the neutral regions, thereby maintaining **charge neutrality**. However, under low-level injection condition, the change in the majority carrier concentration is negligibly small.

The situation on the n-side is quite similar to the example in Section 3.7, where illumination was used to raise the minority carrier concentration at a surface. In this case,

in a forward biased junction, instead of illumination, minority carriers are injected from the p-side to increase the hole concentration at the surface adjacent to the space charge region. As in the previous example, the excess carriers diffuse into the bulk due to the concentration gradient. Neglecting minority carrier drift current (assumption 2), the steady state continuity equation in the n-region is given by (refer to eqn.3.37)

$$\frac{d^2 p_n}{dx^2} - \frac{(p_n - p_{no})}{L_p^2} = 0 \quad \text{.....(4.44)}$$

Since the hole concentration at $x = x_n$ is given by eqn.(4.43a), and the hole concentration reduces due to recombination to the thermal equilibrium value far away from the junction, the boundary conditions for an infinitely long diode are :

(i) $p_n(x_n) = p_{ne} = p_{no} \exp\left(\frac{V}{V_T}\right)$ at $x = x_n$ and

(ii) $p_n(\infty) = p_{no}$ at $x = \infty$

Now solving eqn.(4.44) following the same procedure as in Section 3.7, we have

$$p_n - p_{no} = p_{no}\left[\exp\left(\frac{V}{V_t}\right) - 1\right]\exp\left(-\frac{x - x_n}{L_p}\right) \text{.....(4.45a)}$$

From eqn.(4.45a) we see that the hole concentration in the n-region ($x > x_n$) will decay exponentially away from the junction. Similarly, the electron in the p-region ($x < -x_p$) will also decay exponentially and the profile is given by

$$n_p - n_{po} = n_{po}\left[\exp\left(\frac{V}{V_t}\right) - 1\right]\exp\left(\frac{x + x_p}{L_n}\right) \text{.....(4.45b)}$$

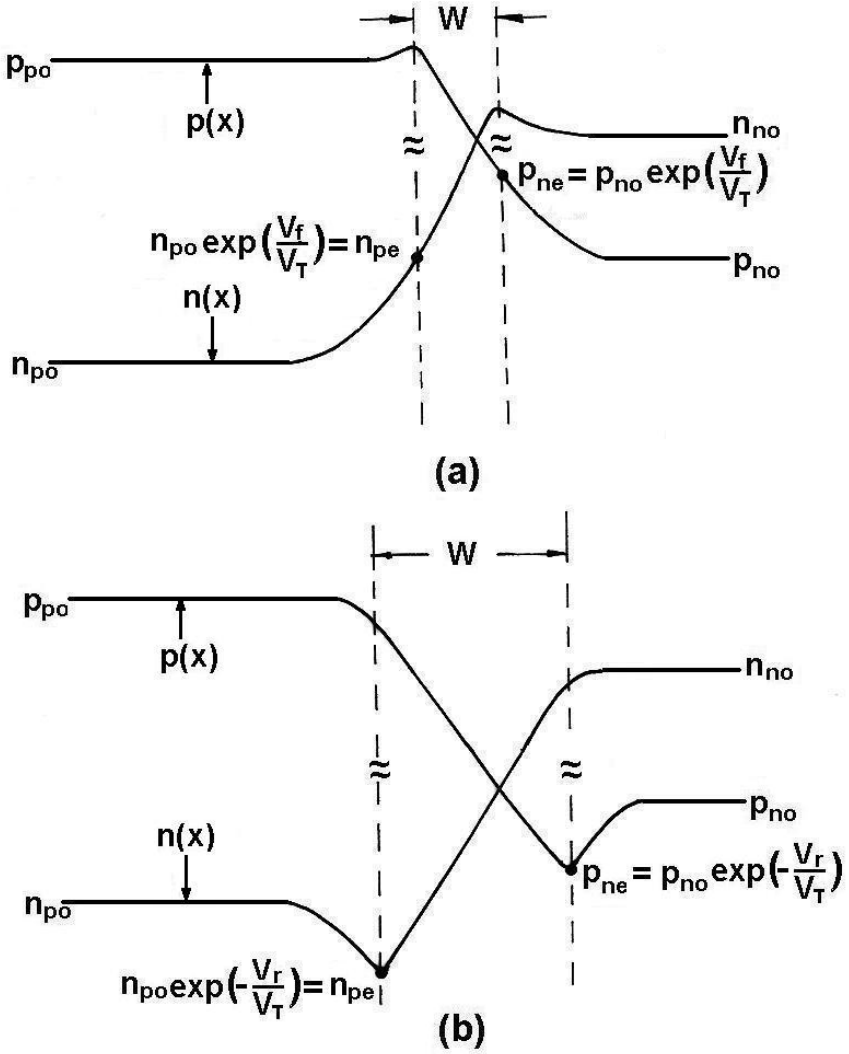


Fig. 4.9 The variation of the electron and hole concentrations in a p-n junction diode for (a) forward bias and (b) reverse biased cases.

Fig. 4.9 shows the variation of the electron and hole concentrations in the diode for both forward and reverse biased cases. Now the hole diffusion current at the edge of the depletion layer is given by

$$I_p(x_n) = -AqD_p \left.\frac{dp_n}{dx}\right|_{x=x_n} = \frac{AqD_p p_{no}}{L_p}\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \quad \text{.....(4.46a)}$$

Similarly, the electron diffusion current at $x = -x_p$ can be obtained as

$$I_n(-x_p) = AqD_n \left.\frac{dn_p}{dx}\right|_{x=-x_p} = \frac{AqD_n n_{po}}{L_n}\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \ldots\ldots..(4.46b)$$

According to our assumption (3), there is no generation or recombination in the depletion region. Therefore the two current components derived above remain constant throughout the depletion layer. The sum of these two components gives the total current flowing through the p-n junction as

$$I = I_p(x_n) + I_n(-x_p) = Aq\left(\frac{D_p p_{no}}{L_p} + \frac{D_n n_{po}}{L_n}\right)\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \ldots\ldots. (4.47)$$

Therefore the diode current can be expressed as

$$I = I_o\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \ldots\ldots..(4.48)$$

where I_o is called the reverse saturation current of the diode and is given by

$$I_o = Aq\left(\frac{D_p p_{no}}{L_p} + \frac{D_n n_{po}}{L_n}\right) = Aqn_i^2\left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A}\right) \ldots\ldots\ldots(4.49)$$

Fig. 4.10 plots the current-voltage characteristics of the diode as given by eqn.(4.48). We see from eqn. (4.48) that when the diode is forward biased (i.e. $V > 0$), the current increases exponentially with applied voltage, while when it is reverse biased (i.e. $V < 0$), the current is limited to $I = -I_o$. This property of the diode is used for several applications.

For a one-sided abrupt p^+n junction $p_{no} >> n_{po}$, and hence in the expression of I_o given by eqn.(4.49), only the first term dominates. Thus, for such a situation, the reverse saturation current can be approximated as

$$I_o = \frac{AqD_p p_{no}}{L_p} = \frac{AqD_p n_i^2}{L_p N_D} \ldots\ldots.(4.50)$$

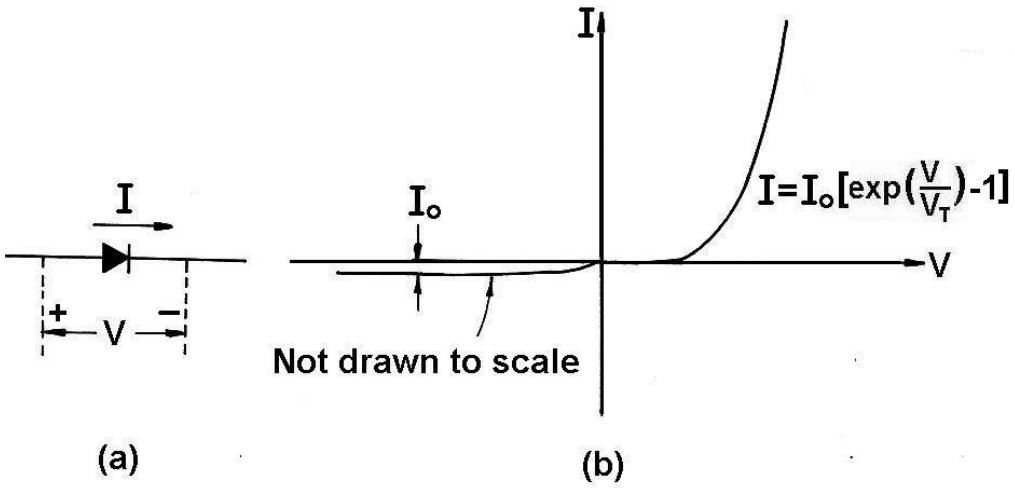


Fig. 4.10 Current-voltage characteristics of a p-n junction diode

An interesting point to be noted is that although the current in a p^+n junction is primarily due to holes injected from the heavily doped p-side into the n-region, the actual magnitude of the current is independent of the doping concentration on the p-side but depends on the doping concentration of the lightly doped n-side. In such a junction, decrease in the donor concentration in the n-region results in an increase in I_o.

Fig.4.11 plots the hole and electron currents flowing through a forward biased diode as a function of x. Using eqn. (4.45), we can write the expressions for the minority carrier diffusion current in the neutral n-region ($x > x_n$) as

$$I_p(x) = -AqD_p \frac{dp_n}{dx} = I_p(x_n)\exp\left(-\frac{x - x_n}{L_p}\right) \quad \text{.....(4.51a)}$$

Similarly, the minority carrier diffusion current in the neutral p-region ($x < -x_p$) is given by

$$I_n(x) = AqD_n \frac{dn_p}{dx} = I_n(-x_p)\exp\left(-\frac{x + x_p}{L_n}\right) \quad \text{.....(4.51b)}$$

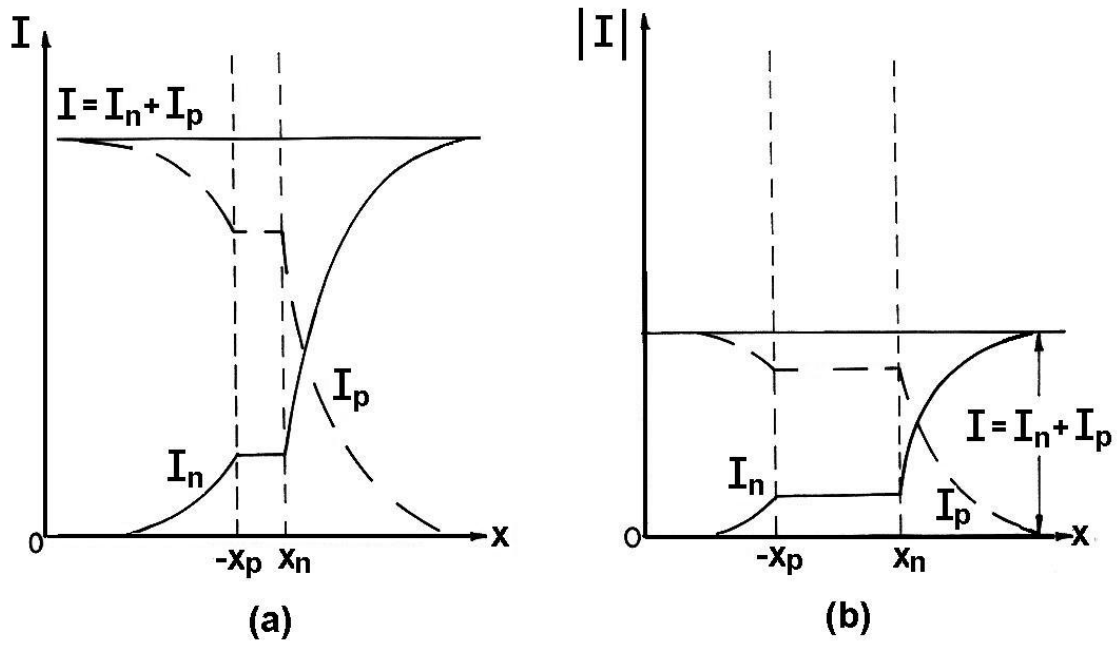


Fig. 4.11 Variation of electron and hole current components in a p-n junction under (a) forward bias and (b) reverse bias conditions

From eqn. (4.51), it is obvious that the components of the minority carrier diffusion current decay exponentially away from the junction. However, since the total current flowing in a diode must remain constant, the decay in the minority carrier current is compensated by a corresponding increase in the majority carrier current as shown in Fig.4.11(a). Thus, far away from the junction, on the n-side, I_p falls to zero and the current is carried entirely by electrons flowing from the n-contact towards the junction. To reiterate, let us trace the flow of electrons from the n-contact to the p-contact. Electrons flowing in at the n-contact from the negative terminal of the battery move towards the junction as a drift current with the aid of a small electric field in the neutral n-region. The field required for this flow is very small as electrons are the majority carriers and has therefore been neglected in our analysis by assumption (2). As the electrons flow towards the junction, they partially recombine with the holes moving in

the opposite direction resulting in a decrease in electron current. (Since electrons and holes recombine in pairs, the decrease in electron current is equal to the decrease in hole current in the n-region. This ensures that the sum of the electron and hole current is always constant). The electrons, which reach the junction, are injected into the p-side as a minority carrier diffusion current. Far away from the junction, all the electrons recombine, reducing the electron current to zero. Similarly, at the p-contact, the entire current is carried by holes, and this current ultimately reduces to zero as the hole moves across the junction towards the n-contact.

Fig. 4.11(b) shows the variation of electron and hole current densities for a reverse biased diode. In this case, since the minority carrier concentrations in the neutral regions are reduced below the thermal equilibrium values, there is excess generation of carriers. The electrons, which are generated on the p-side of the junction, diffuse towards the junction and are swept across it by the electric field in the space charge regions. The electron current increases further in the n-side due to generation, as these carriers flow towards the positive terminal of the battery connected to the n-contact. Similarly, the holes flow from the n-region to the p-region.

Physically, one can now understand why the forward current is much larger than the reverse bias current. Under forward bias conditions, the hole current injected from the p-side, where holes are majority carriers, into the n-side is proportional to $p_{no}\left[\exp\left(\frac{V}{V_T}\right)-1\right]$, which can be very large. On the other hand, under reverse bias conditions, holes are injected from the n-side, where they are minority carriers, to the p-side. The current, in this case, depends on the generation rate of carriers, which is

proportional to p_{no}. This explains why the reverse current is almost independent of the reverse bias voltage and is equal to I_o, while the forward bias current is higher by a factor $\left[\exp\left(\frac{V}{V_T}\right)-1\right]$.

Q 4.2 Although the excess holes in the neutral n-region are continuously recombining, how is it that the steady state minority carrier concentration profile shown in Fig.4.9 do not change with time?

Let us at first calculate the total minority carrier recombination rate, i.e. the number of holes recombining per unit time, in the neutral n-region. Let (Q_p/q) be the total excess hole concentration in the neutral n-region. Now since τ_p is the average lifetime of these holes, the total recombination rate (R) is given by $(Q_p/q\tau_p)$, or we can write

$$R = \frac{Q_p}{q\tau_p} = \frac{A}{\tau_p}\int_{x_n}^{\infty}(p_n - p_{no})dx = \frac{A}{\tau_p}\int_{x_n}^{\infty} p_{no}\left[\exp\left(\frac{V}{V_t}\right)-1\right]\exp\left(-\frac{x-x_n}{L_p}\right)dx$$
$$= \frac{Ap_{no}L_p}{\tau_p}\left[\exp\left(\frac{V}{V_t}\right)-1\right]$$

On the other hand, the injected hole current is given by

$$I_p(x_n) = \frac{AqD_p p_{no}}{L_p}\left[\exp\left(\frac{V}{V_T}\right)-1\right] = \frac{AqD_p\tau_p p_{no}}{L_p\tau_p}\left[\exp\left(\frac{V}{V_T}\right)-1\right] = \frac{Aqp_{no}L_p}{\tau_p}\left[\exp\left(\frac{V}{V_t}\right)-1\right]$$

Thus we see that the number of holes injected into the n-region per unit time given by $I_p(x_n)/q$ is exactly equal to the number of holes recombining per unit time, thereby maintaining the steady state profile. A similar analysis can also be carried out for electrons in the p-region.

4.3.2 Quasi-Fermi levels under bias condition

With this background let us now refer to Fig.4.6 and take another look at the energy band diagram with application of bias. As there is no change in the majority carrier concentration, the quasi Fermi level for holes (E_{Fp}) deep inside the neutral p-region and the quasi-Fermi level for electrons (E_{Fn}) deep inside the neutral n-region are essentially at the equilibrium value and remain constant. However, at the edges of the depletion layer, the minority carrier concentration changes from its equilibrium value by a factor $\exp\left(\frac{V}{V_T}\right)$, so that at the edges of the depletion region $p.n = n_i^2 \exp\left(\frac{V}{V_T}\right)$. Referring to eqn.(1.51), this implies a separation between E_{Fp} and E_{Fn} on either side of the junction by an amount qV, as shown in Fig.4.6(a) and (b). It can be easily seen that under forward bias, the quasi-Fermi level for electrons (E_{Fn}) is above that for holes (E_{Fp}) by qV_f, and under reverse bias, E_{Fp} is above E_{Fn} by qV_r. As the minority carriers decay away from the depletion layer edge towards the bulk and approach their equilibrium values, E_{Fp} and E_{Fn} come closer and ultimately merge. Also, in the absence of significant generation-recombination, E_{Fp} and E_{Fn} remain constant throughout the depletion region.

4.3.3 Current-Voltage relation in practical diodes having finite length

For a practical diode with finite lengths, the boundary conditions change and the equations derived in the previous section have to be modified. The continuity equation given by eqn.(4.44) will now have to be solved subject to these new boundary conditions. While the first boundary condition specifying the carrier concentration at the edge of the depletion region remains the same, the second boundary condition has to be changed to take into account the finite length of the neutral regions. We have already discussed the role of surface recombination in Section 1.4.4. The surface recombination velocities at

the metal-semiconductor contacts are very high resulting in high recombination rates. Consequently, the excess carrier concentration reduces to zero at these contacts. The boundary conditions for the solution of eqn.(4.44) in the case of a diode with a n-region of finite length are therefore

(i) $p_n(x_n) = p_{ne} = p_{no}\exp\left(\frac{V}{V_T}\right)$ at $x = x_n$ and

(ii) $p_n(W_1) = p_{no}$ at $x = W_1$

where $W_1 = x_n + W_n$, W_n being the width of the neutral n-region. The solution of eqn.(4.44) is now of the form

$$p_n - p_{no} = p_{no}\left[\exp\left(\frac{V}{V_t}\right) - 1\right]\frac{\mathrm{Sinh}\left(\frac{W_1 - x}{L_p}\right)}{\mathrm{Sinh}\left(\frac{W_1 - x_n}{L_p}\right)} \quad \ldots.(4.52)$$

Eqn.(4.52) shows the variation of the minority carrier distribution in the n-region. From eqn.(4.52), it can be shown that when $(W_n/L_p) >> 1$, i.e. the width of the n-layer is much larger than the diffusion length, the expression for $(p_n - p_{no})$ reduces to that of eqn.(4.45) for an infinitely long n-region, indicating an exponential decay of carriers away from the junction. A very interesting situation is encountered for a short diode. For $y << 1$, $\mathrm{Sinh}(y) \approx y$. So for $W_n << L_p$, eqn. (4.52) reduces to

$$p_n - p_{no} = \frac{p_{no}(W_1 - x)}{W_n}\left[\exp\left(\frac{V}{V_t}\right) - 1\right] \quad \ldots.(4.53)$$

In other words, the minority carrier concentration approaches a linear distribution instead of an exponential distribution. The nature of variation of the minority carrier concentration for various (W_n/L_p) values is shown in Fig. 4.12.

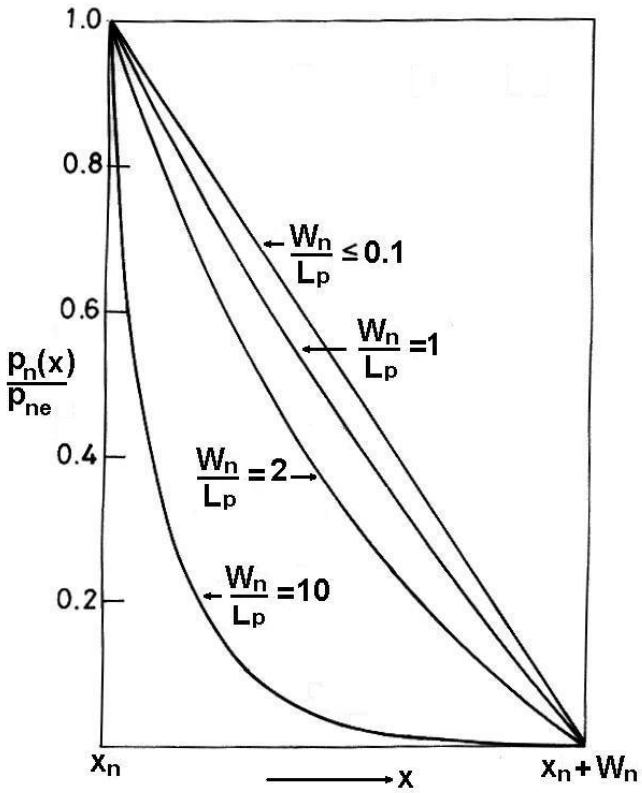


Fig. 4.12 Hole concentration variation in the neutral n-region of a p-n junction for various (W_n/L_p) ratios

The hole diffusion current in the neutral n-region can now be expressed as

$$I_p(x) = -AqD_p \frac{dp_n}{dx} = \frac{AqD_p p_{no}}{L_p}\left[\exp\left(\frac{V}{V_t}\right) - 1\right]\frac{\text{Cosh}\left(\frac{W_1 - x}{L_p}\right)}{\text{Sinh}\left(\frac{W_1 - x_n}{L_p}\right)} \quad \ldots\ldots(4.53)$$

So, the hole diffusion current at the edge of the depletion layer on the n-side is given by

$$I_p(x_n) = -AqD_p \left.\frac{dp_n}{dx}\right|_{x=x_n} = \frac{AqD_p p_{no}}{L_p}\left[\exp\left(\frac{V}{V_t}\right) - 1\right]\text{Coth}\left(\frac{W_n}{L_p}\right) \quad \ldots..(4.54a)$$

Similarly, if the p-region is also of finite length, the electron diffusion current at the edge of the depletion layer on the p-side is given by

$$I_n(-x_p) = \frac{AqD_n n_{po}}{L_n}\left[\exp\left(\frac{V}{V_t}\right)-1\right]\mathrm{Coth}\left(\frac{W_p}{L_p}\right) \quad \text{.....(4.54b)}$$

where W_p is the width of the neutral p-region. The total diode current is now given by adding these two components as

$$I = Aq\left[\frac{D_p p_{no}\mathrm{Coth}\left(\frac{W_n}{L_p}\right)}{L_p} + \frac{D_n n_{po}\mathrm{Coth}\left(\frac{W_p}{L_n}\right)}{L_n}\right]\left[\exp\left(\frac{V}{V_t}\right)-1\right]$$

$$= Aqn_i^2\left[\frac{D_p\mathrm{Coth}\left(\frac{W_n}{L_p}\right)}{N_D L_p} + \frac{D_n\mathrm{Coth}\left(\frac{W_p}{L_n}\right)}{N_A L_n}\right]\left[\exp\left(\frac{V}{V_t}\right)-1\right] \quad \text{.....(4.55)}$$

Now let us look at two extreme cases:

Case 1. Long Diode (also called **long-base diode**): For y > 4, Coth(y) ≈ 1. Therefore if $W_n > 4L_p$ and $W_p > 4L_n$, eqn. (4.55) will reduce to the ideal diode equation given by eqn. (4.47). So the ideal diode equation is said to be valid for a long diode satisfying the above restrictions regarding the widths of the n and p-regions.

Case 2. Short Diode (also called **short-base diode**): We have already seen that the minority carrier concentration profile in the neutral region of a short diode shows a linear variation. Since the slope of this profile is constant, the minority carrier diffusion current in the neutral region also remains constant. Since Cosh(y) ≈ 1, when y is small, it can be seen from eqn.(4.53) that $I_p(x)$ is a constant when $W_n << L_p$. This is to be expected as $W_n << L_p$ implies that there will be very little recombination in the neutral region, since the width of the device is much less than the diffusion length.

Since Coth(y) ≈ (1/y), when y is small, from eqn. (4.55) we have

$$I = Aqn_i^2\left[\frac{D_p}{N_D W_n} + \frac{D_n}{N_A W_p}\right]\left[\exp\left(\frac{V}{V_t}\right) - 1\right] = I_o\left[\exp\left(\frac{V}{V_t}\right) - 1\right] \quad \text{.....(4.56a)}$$

where

$$I_o = Aqn_i^2\left[\frac{D_p}{N_D W_n} + \frac{D_n}{N_A W_p}\right] = Aq\left[\frac{D_p p_{no}}{W_n} + \frac{D_n n_{po}}{W_p}\right] \quad \text{.....(4.56b)}$$

We also see that eqn.(4.56) is similar to eqn.(4.47) derived for an infinitely long diode except that the diffusion lengths (L_p and L_n) are replaced by the widths of the neutral regions (W_n and W_p). This is because $\left.\frac{dp_n}{dx}\right|_{x=x_n} = -\frac{p_n'}{L_p}$ for a long base diode, while $\left.\frac{dp_n}{dx}\right|_{x=x_n} = -\frac{p_n'}{W_n}$ for a short base diode. The electron and hole currents flowing through the diode as a function of x are now plotted in Fig.4.13. This property of a short base diode will prove particularly useful in our discussion of Bipolar Junction Transistors in Chapter 6.

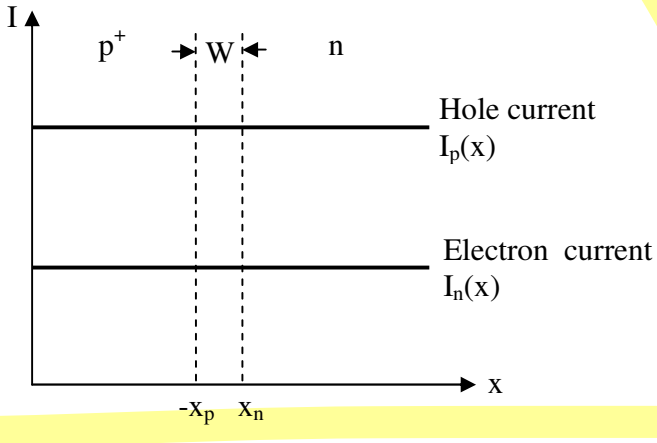


Fig. 4.13 Electron and hole current variation in a p-n junction diode with short p- and n-regions

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Q. 4.3 We have seen (refer to Q.4.2) that $I_p(x=x_n) = Q_p/\tau_p$ in the neutral n-region of a long base diode. Is this also valid for a short base diode?

For a long base diode, all the holes injected into the n-region recombine before reaching the n-contact, and Q_p/τ_p represents the recombination current in the neutral base region. However, in the case of a short base diode, a hole current exists even at the n-contact, signifying a large concentration of holes together with the majority carrier electrons at the n-contact. Since surface recombination velocities are extremely large, there will be a large recombination current at the n-contact surface. Since the holes recombining at the surface, in addition to those recombining in the bulk, given by Q_p/τ_p, are supplied by the injected hole current, $I_p(x=x_n)$ will not be equal to Q_p/τ_p in a short base diode. It has been left as a problem at the end of the chapter to show that $I_p(x=x_n) > Q_p/\tau_p$ for a short base diode.

Example 4.5: For an abrupt p^+n silicon diode, the doping concentration on the n-side is 10^{16} cm^{-3}. The width of the n-region is 2 μm. Assuming this width is much smaller than the hole diffusion length in the n-region, calculate the reverse saturation current at (i) 300K and (ii) 400K if the area of the diode is 100μm x 100 μm. Assume that the hole mobility in the n-region is 350cm^2/V-sec at both the temperatures.

Solution: From eqn. 4.56, we know that for a short diode, $I_o = Aqn_i^2\left[\frac{D_p}{N_D W_n} + \frac{D_n}{N_A W_p}\right]$.

In a p^+n diode, since $N_A >> N_D$, $I_o \approx Aqn_i^2\left[\frac{D_p}{N_D W_n}\right]$

Also we know from Einstein's relation, $D_p = \mu_p V_T$

At 300K, $D_p = 0.026\text{x}350\ \text{cm}^2/\text{sec} = 9.1\ \text{cm}^2/\text{sec}$ and $n_i = 1.5\text{x}10^{10}\ \text{cm}^{-3}$

$$\therefore I_o = \frac{\left(100\text{x}10^{-4}\right)^2 \text{x}1.6\text{x}10^{-19}\,\text{x}9.1\text{x}\left(1.5\text{x}10^{10}\right)^2}{2\text{x}10^{-4}\,\text{x}10^{16}} = 1.638\text{x}10^{-14}\,\text{A}$$

At 400K, $D_p = 0.035\text{x}350\ \text{cm}^2/\text{sec} = 12.25\ \text{cm}^2/\text{sec}$ and $n_i = 5.185\text{x}10^{12}\ \text{cm}^{-3}$ (see Ex. 1.1)

$$\therefore I_o = \frac{\left(100\text{x}10^{-4}\right)^2 \text{x}1.6\text{x}10^{-19}\,\text{x}12.25\text{x}\left(5.185\text{x}10^{12}\right)^2}{2\text{x}10^{-4}\,\text{x}10^{16}} = 2.635\text{x}10^{-9}\,\text{A}$$

So, we see that for100°C rise in temperature, the reverse saturation current increases by five orders of magnitude!

Q.4.4: Compare the plots of the forward current-voltage characteristics of Ge, Si and GaAs p^+n diodes having identical structures and doping concentrations.

For two materials, material1 and material2, having band gap energies of E_{g1} and E_{g2}, the ratio of their intrinsic carrier concentrations n_{i1} and n_{i2} are given by (from eqn.1.24)

$$\frac{n_{i1}}{n_{i2}} = \frac{\sqrt{N_{C1}N_{V1}}\exp\left[-\left(\frac{E_{g1}}{2kT}\right)\right]}{\sqrt{N_{C2}N_{V2}}\exp\left[-\left(\frac{E_{g2}}{2kT}\right)\right]} = \sqrt{\frac{N_{C1}N_{V1}}{N_{C2}N_{V2}}}\exp\left(\frac{E_{g2}-E_{g1}}{2kT}\right)$$

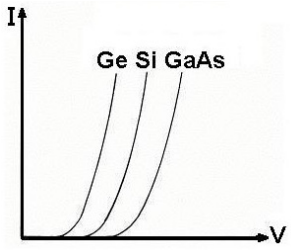
Assuming that the differences in the values of N_C and N_V are small,

$$\frac{n_{i1}^2}{n_{i2}^2} \approx \exp\left(\frac{E_{g2}-E_{g1}}{kT}\right)$$

Considering the bandgaps given in Table 1.3, $\frac{(n_i^2)_{Ge}}{(n_i^2)_{Si}} \approx \exp\left(\frac{1.12-0.66}{0.026}\right) = 4.83\text{x}10^7$

Similarly, $\frac{(n_i^2)_{Si}}{(n_i^2)_{GaAs}} \approx \exp\left(\frac{1.42-1.12}{0.026}\right) = 1.03\text{x}10^5$

Since other terms in the expression for I_o have much smaller dependence on the material, we may say, $(I_o)_{Ge} >> (I_o)_{Si} >> (I_o)_{GaAs}$ resulting in the I-V characteristics shown below.



4.3.4 Ideality factor of a p-n junction diode

We have seen in the preceding section that the current in a p-n junction diode can be expressed by eqn. (4.48). For sufficiently large forward bias voltage ($V >> V_T$), the diode current can be further approximated as

$$I \approx I_o \exp\left(\frac{V}{V_T}\right) \ldots\ldots\ldots (4.57)$$

In practice, however, the characteristics of a real diode follows a slightly modified equation given by

$$I \approx I_o \exp\left(\frac{V}{mV_T}\right) \ldots\ldots\ldots (4.58)$$

where m is called the ideality factor of the diode and its value lies between 1 and 2. This difference between a real diode and an ideal diode is present because we have neglected the recombination within the space-charge layer (Assumption 3) in our analysis. In a real diode, because of recombination within the depletion layer, there is another component of current given by[1]

$$I_{rec} = \frac{Aqn_i W}{2\tau} \exp\left(\frac{V}{2V_T}\right) \ldots\ldots (4.59)$$

Thus, for a real p^+n junction under forward bias, the diode current will actually be the sum of I_{rec} (given by eqn. (4.59)) and I (given by equation (4.57)). The ratio I_{rec}/I signifying the relative dominance of these two currents will determine the value of m. For an abrupt p^+n junction, from eqns. (4.48), (4.50) and (4.59) we have

$$\frac{I_{rec}}{I} = \frac{WN_D}{2L_p n_i} \exp\left(-\frac{V}{2V_T}\right) \ldots\ldots (4.60)$$

We see from eqn. (4.60) that for large forward biases, i.e. $V >> (2V_T)$, $I_{rec} << I$, and therefore $m \approx 1$. However, when the forward bias is reduced, the semiconductor material properties as well as the operating temperature play an important role in determining this ratio. For example, in a germanium diode n_i is large ($2.4 \times 10^{13}/cm^3$ at 300K) and hence I_{rec}/I is small. The diode current is therefore dominated by the diffusion current and $m \approx 1$. On the other hand, for GaAs, n_i is small ($1.79 \times 10^6/cm^3$ at 300K), therefore I_{rec}/I is large and $m \approx 2$. For silicon diodes, at room temperature, up to a moderate forward bias of 0.4 – 0.5V, I_{rec} dominates and m is close to 2. Fig.4.14 shows the nature of variation of the diode current and ideality factor as a function of the applied forward voltage for p-n junction diodes using Ge, Si and GaAs.

4.4 Transient analysis

So far we have discussed about the p-n junction in steady state. We have seen that any change in the diode current will lead to a change in the charge distribution. However, it is evident that the distribution of carriers cannot change instantaneously and some time is required for increasing the concentration or reducing it. Thus the stored charge must inevitably lag behind the current. This is inherently a capacitive effect, which will be discussed in this Section.

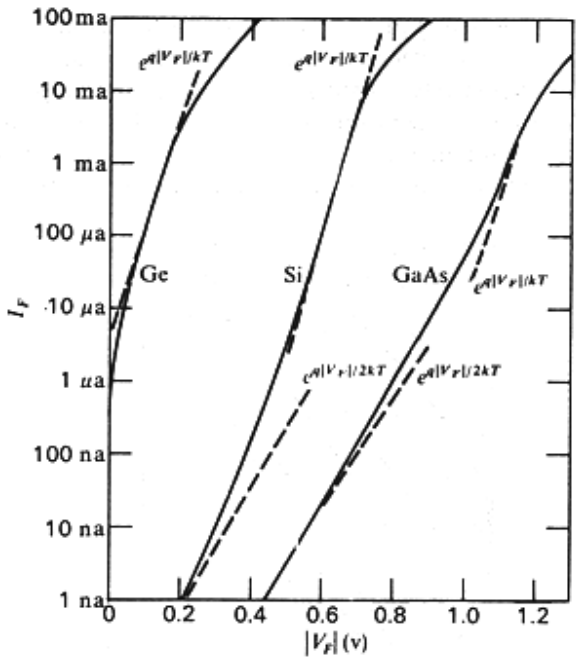


Fig.4.14 Forward bias current-voltage characteristics of Germanium, Silicon and Gallium Arsenide p-n junction diodes[1] (From A.S. Grove, copyright © 1967 by John Wiley & Sons, Inc. Used by permission)

4.4.1 Time Variation of Stored Charge

Let us consider a p^+n diode with a long n-region. In order to analyze the transient behaviour of this diode, we have to solve the time-dependent continuity equation given by eqn. (3.30b). From this we get

$$-\frac{\partial J_p}{\partial x} = q\left[\frac{\partial p_n}{\partial t} + \frac{p_n - p_{no}}{\tau_p}\right] \ldots\ldots(4.61)$$

where J_p and p_n are functions of both x (position) and t (time). For a p^+n junction diode with a long n-region, the total current can be approximated by the injected hole current into the neutral n-region at $x = x_n$, i.e. $I(t) \approx AJ_p(x_n,t)$, where A is the cross-sectional area of the diode. On the other hand at $x = \infty$, the hole current is zero, i.e. $J_p(\infty,t) = 0$. Then, from eqn.(4.61), we get the total diode current as a function of time given by

$$I(t) = A\int_{x_n}^{\infty}\left(-\frac{\partial J_p}{\partial x}\right)dx = A\left[J_p(x_n,t) - J_p(\infty,t)\right] = Aq\int_{x_n}^{\infty}\left[\frac{p_n - p_{no}}{\tau_p} + \frac{\partial p_n}{\partial t}\right]dx$$
$$= \frac{Q_p(t)}{\tau_p} + \frac{\partial Q_p(t)}{\partial t} \qquad \text{.....(4.62)}$$

where $Q_p = Aq\int_{x_n}^{\infty}(p_n - p_{no})dx$ is the excess hole charge stored in the neutral n-region of the diode.

Physically, eqn. (4.62) states that the hole current injected across the p^+n junction (and therefore, approximately the total diode current) is related to two charge storage effects. The first is the usual recombination term Q_p/τ_p. The second term is $\partial Q_p/\partial t$, which represents the fact that the distribution of excess charge can increase or decrease with time. In steady state, when $\partial Q_p/\partial t = 0$, the injected current is equal to the recombination rate. However, injection of holes may result in an increase in the excess hole charges stored in the neutral n-region, if the rate of inflow (i.e. the injected current) is greater than the recombination rate. On the other hand, there will be a reduction in the stored charge (i.e. $\partial Q_p/\partial t$ will be negative), if the rate of inflow is less than the recombination rate. Also, the rate of increase of charge is equal to the rate at which charges flow in minus the rate at which the carriers recombine, or more simply, if X be the rate at which carriers flow in and Y be the rate at which carriers recombine per unit time, then X – Y is the rate of increase in the number of carriers.

Now, for a given current transient, the stored charge can be obtained as a function of time using eqn. (4.62). The following example will clarify the concept. Let us assume that a constant current I is flowing through a diode. Since the diode is at steady state,

$\frac{\partial Q_p}{\partial t} = 0$, and from eqn.(4.62), we have $Q_p = I\tau_p$. At time t = 0, the connection to the battery is open-circuited so that the current suddenly becomes zero as shown in Fig. 4.15(a). Since the excess holes in the n-region must die out by recombination, some time is needed for Q_p to become zero. Eqn.(4.62) now reduces to

$$\frac{\partial Q_p}{\partial t} = -\frac{Q_p(t)}{\tau_p} \quad \ldots\ldots(4.63)$$

Solving eqn. (4.63) using the initial condition $Q_p(0) = I\tau_p$, we get

$$Q_p(t) = I\tau_p \exp\left(-\frac{t}{\tau_p}\right) \quad \ldots.(4.64)$$

In other words, as shown in Fig.4.15(b), the stored charge dies exponentially with a time constant τ_p, i.e. the hole lifetime in the n-region.

An important point to be noted in the above example is that although the diode current has suddenly fallen to zero, the voltage across the junction does not immediately do so. Since the excess hole concentration at the edge of the depletion region p'_{ne} is related to the junction voltage, we can write using equation 4.43(a)

$$p'_{ne} = p_{no}\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \quad \ldots..(4.65)$$

Therefore from eqn. (4.65), we have

$$V(t) = V_T \ln\left[\frac{p'_{ne}(t)}{p_{no}} + 1\right] \ldots\ldots(4.66)$$

If we know $p'_{ne}(t)$, we can find the junction voltage as a function of time. Unfortunately, it is not so simple. Since I = 0, the slope of $p'_n(x)$ must be zero at $x = x_n$ for $t > 0$ (refer to

eqn.(4.46a)) and $p'_n(x,t)$ has the shape as shown in Fig. 4.15(c). Therefore $p'_n(x)$ no longer varies exponentially with x. However, neglecting this minor variation and writing the stored charge at any instant as given by eqn.(4.45), we have

$$Q_p(t) = \int_{x_n}^{\infty} p'_{ne}(t)\exp\left(-\frac{x - x_n}{L_p}\right)dx = AqL_p p'_{ne}(t) \quad \ldots\ldots(4.67)$$

We know that $Q_p(t)$ reduces exponentially as given by eqn.(4.64). Using eqns.(4.64) and (4.67) to find an expression for $p'_{ne}(t)$, and substituting this expression in eqn.(4.66), we have

$$V(t) = V_T \ln\left[\frac{I\tau_p}{AqL_p p_{no}}\exp\left(-\frac{t}{\tau_p}\right) + 1\right] \quad \ldots\ldots.(4.68)$$

Therefore, it is easily understood that if the junction voltage has to fall to zero quickly, τ_p must be small. This can be achieved by deliberately introducing gold or platinum as impurities as discussed in Section 1.4.3. Au and Pt create available states near the middle of the bandgap. This helps the recombination process and reduces the minority carrier lifetime τ_p. Alternately, for fast switching diodes, the device geometry may also be modified by designing a p^+n diode with a very narrow base (n-region). If the n-region width is less that the hole diffusion length, very few charges can be stored and hence a very short time is required to switch the diode off and on.

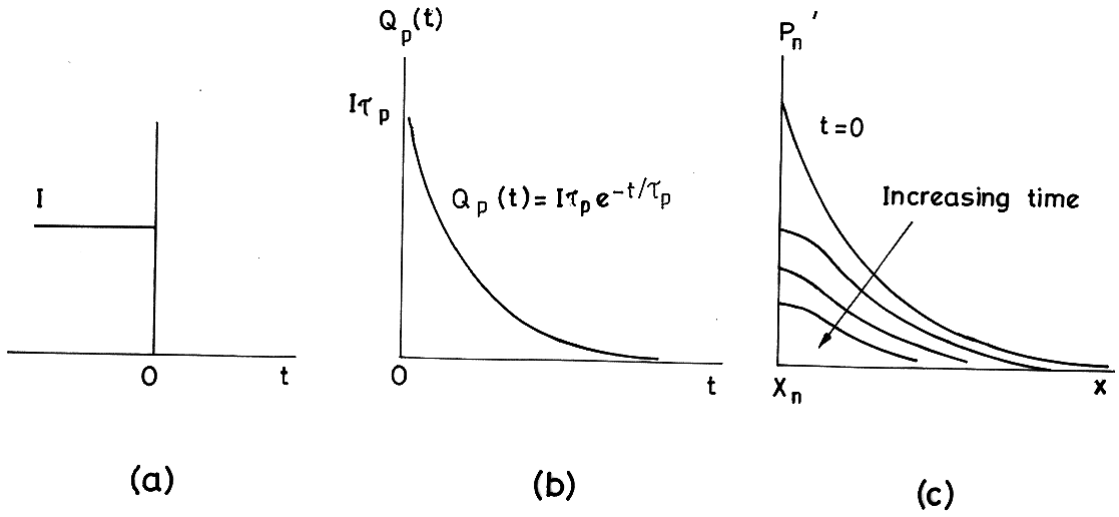


Fig.4.15 (a) The time variation of current flowing through a p^+n junction diode and the resultant (b) stored charge variation and (c) charge distribution variation in the neutral n-region

4.4.2 Reverse Recovery of a diode

In most switching applications, the diode has to be switched from forward conduction to reverse-biased state and back. As we shall see, in such a situation, a reverse current much larger than the reverse saturation current (I_o) can flow through the diode for a short time. Let us consider the diode circuit shown in Fig. 4.16(a) and assume that the input voltage (V_a) is varied as shown in fig. 4.16(b). At $t < t_1$, $V_a = V_F$ and in steady state, the current I_F flowing through the circuit is limited by the resistor R. Neglecting the small forward voltage drop of the diode (V_d), $I = I_F \approx (V_F - V_d)/R \approx V_F/R$. At $t > t_1$, the applied voltage becomes $-V_R$. However, as already pointed out, the stored charge and hence the junction voltage (V_d) cannot change immediately as shown in Figs. 4.16(c) and 4.16(e). So just as the voltage is reversed, the diode junction voltage remains at the same small forward bias value. Consequently, a large reverse current $I = -I_R = (-V_R - V_d)/R \approx -V_R/R$

flows temporarily as shown in fig. 4.16(d). With time, as the stored charge is depleted, the junction voltage reduces. So long as Q_p is positive, the junction voltage has a small positive value. The current remains approximately constant at $I = -V_R/R$ till Q_p falls to zero. However, once Q_p becomes negative, the junction voltage also becomes negative. As the source voltage begins to be divided between the diode and the resistor, the magnitude of the current reduces. Finally, almost all the applied voltage is dropped across the diode and the current in the circuit becomes equal to I_o. The time required for Q_p and the junction voltage to become zero is called the storage delay time (t_{sd}). Evidently, t_{sd} depends on τ_p since the rate of removal of stored charge depends on the carrier lifetime.

We shall now obtain an expression for t_{sd} in terms of τ_p and the diode current by solving eqn.(4.62) for this particular situation. We note that at $t = t_1$, the diode current becomes $-I_R$ and remains constant till $t = t_1 + t_{sd}$. Substituting in eqn. (4.62), for $0 < t < t_{sd}$, we have

$$-I_R = \frac{Q_p(t)}{\tau_p} + \frac{dQ_p(t)}{dt} \quad \ldots\ldots(4.69)$$

The solution of this equation is of the form $Q_p(t) = C_1 \exp\left(-\frac{t}{\tau_p}\right) - I_R \tau_p$, where C_1 is the constant of integration to be obtained from the initial condition. We know that since a constant current I_F was flowing through the diode till the time $t = t_1$, the stored charge Q_p at $t = t_1$ is $I_F\tau_p$. Thus we can write the solution of eqn. (4.69) as

$$Q_p(t) = I_F \tau_p \exp\left(-\frac{t - t_1}{\tau_p}\right) + I_R \tau_p \left[\exp\left(-\frac{t - t_1}{\tau_p}\right) - 1\right] \quad \ldots\ldots(4.70)$$

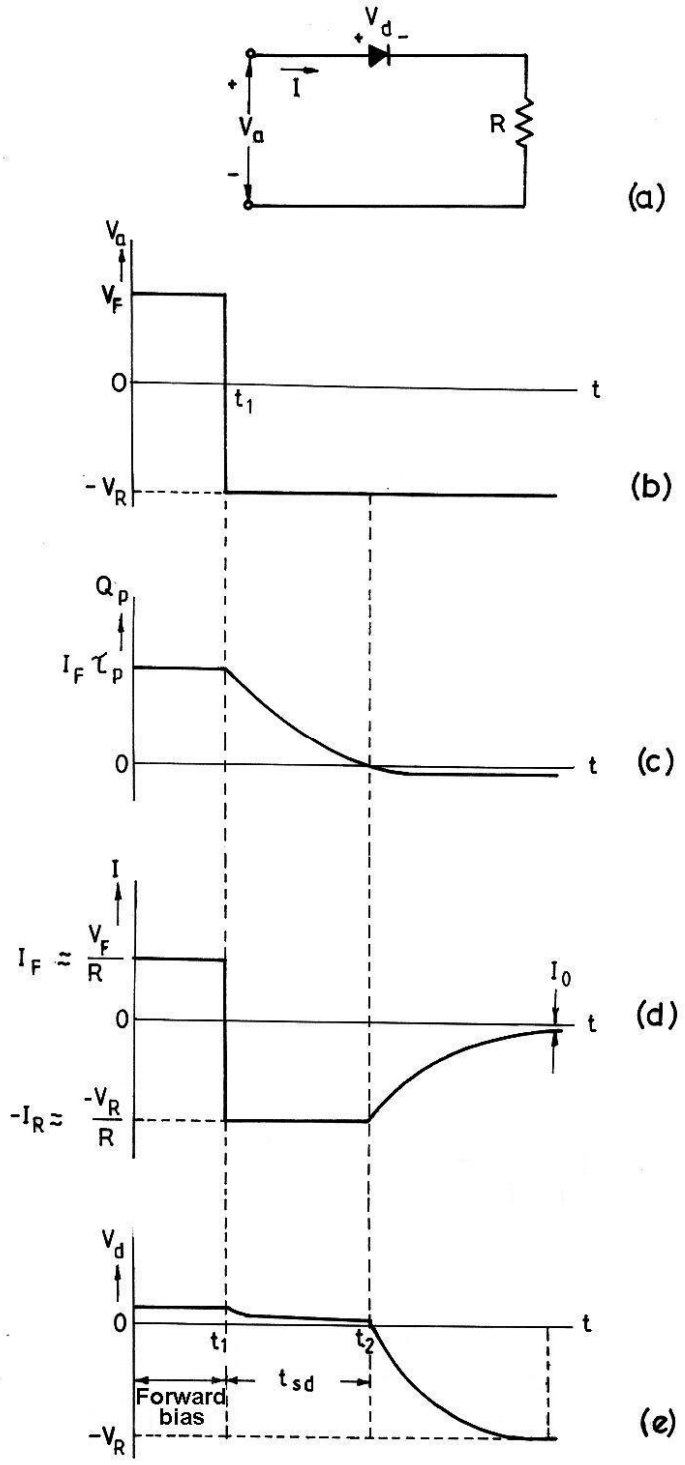


Fig. 4.16 (a) A diode circuit with (b) the input voltage variation and the corresponding variation in (c) minority carrier stored charge in the neutral region, (d) diode current and (e) voltage drop across the diode

Now, substituting eqn.(4.67) in eqn. (4.70), we have

$$p'_{ne}(t) = \frac{1}{AqL_p}\left[(I_F + I_R)\tau_p \exp\left(-\frac{t - t_1}{\tau_p}\right) - I_R\tau_p\right] \ldots\ldots\ldots(4.71)$$

By our definition of the storage delay time, $p'_{ne}(t) = 0$ at $t = t_1 + t_{sd}$. Therefore, from eqn. (4.71), we get

$$t_{sd} = \tau_p \ln\left(1 + \frac{I_F}{I_R}\right) \ldots\ldots(4.72)$$

Thus we see that the presence of a large reverse current reduces t_{sd}. This is to be expected, since the reverse current helps to remove the excess stored charge and I_R is the rate at which excess holes are removed. Also, t_{sd} is higher for higher I_F, as in that case the initial stored charge itself is more.

4.4.3 Charge Storage Capacitance

There are essentially two capacitances associated with a junction. One of them is due to the charge dipole at the depletion layer (Depletion capacitance) and has already been discussed in Sections 4.2.1 and 4.2.2. In addition, as pointed out already, the junction voltage lags behind the diode current due to charge storage and therefore we have another capacitance effect. This is termed the **charge storage capacitance** or the **diffusion capacitance** (C_d). While the depletion capacitance dominates when the diode is in reverse biased condition, the diffusion capacitance becomes important when the junction is forward biased. To calculate this diffusion capacitance due to charge storage effects, let us consider a p^+n junction with a long n-region. Also, let the diode be forward biased with a steady current I flowing through it. Assuming that the junction voltage $V > 3V_T$, from eqn.(4.67) the stored charge can be given as

$$Q_p = I\tau_p = \int_{x_n}^{\infty} p'_{ne} \exp\left(-\frac{x - x_n}{L_p}\right) dx = AqL_p p'_{ne} \approx AqL_p p_{no} \exp\left(\frac{V}{V_t}\right) \ldots\ldots (4.73)$$

The diffusion capacitance (C_d) is therefore given by

$$C_d = \frac{dQ}{dV} = \frac{AqL_p p_{no}}{V_T} \exp\left(\frac{V}{V_t}\right) = \frac{I\tau_p}{V_T} \ldots\ldots (4.74)$$

The small signal conductance of the diode $\left(G = \frac{dI}{dV}\right)$, calculated by differentiating the ideal diode current equation (eqn. (4.48)), is given by

$$G = \frac{I}{V_T} \ldots\ldots (4.75)$$

The diffusion capacitance can therefore be expressed also as

$$C_d = G\tau_p \ldots\ldots (4.76)$$

An important point to note here is that C_d is directly proportional to the diode current.

The above equations are valid for long base diode. Let us now consider a p^+n diode with a short n-region. We have already seen that for such a diode, the minority carrier concentration in the n-region shows a linear variation, as shown in Fig. 4.17. From Fig. 4.17, the excess minority carrier charge can be obtained from the shaded area under the profile given by

$$Q_p = \frac{AqW_n p'_{ne}}{2} \ldots\ldots (4.77)$$

while the current $I_p(x_n)$ is obtained from the slope of the profile as

$$I_p(x_n) = -AqD_p \left.\frac{dp_n}{dx}\right|_{x=x_n} = \frac{AqD_p p'_{ne}}{W_n} \ldots\ldots (4.78)$$

Now from eqns.(4.77) and (4.78), we can write for a p^+n diode

$$I \approx I_p(x_n) = \frac{Q_p}{\tau_t} \text{ .....(4.79)}$$

where $\tau_t = \frac{W_n^2}{2D_p}$ is called the **transit time** of the diode. Using $Q_p = I\tau_t$ instead of $Q_p = I\tau_p$ in eqn.(4.74), we get for a short base diode

$$C_d = \frac{I\tau_t}{V_T} = G\tau_t \text{ .....(4.80)}$$

In fact, eqn.(4.80) is the equation for the diffusion capacitance for a diode where $\tau_t = \tau_p$ for a long base diode and $\tau_t = \frac{W_n^2}{2D_n}$ for a short-base diode.

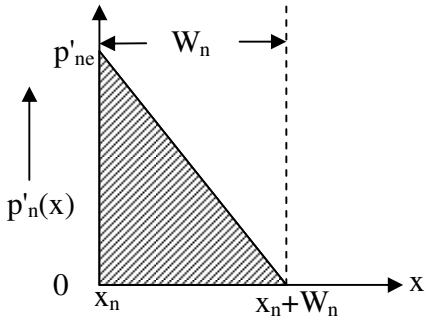


Fig.4.17 Variation of the minority carrier stored charge in the neutral n-region of a short-base p^+n diode

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Q 4.5 Why is τ_t called the transit time?

Let us consider the minority carrier profile in Fig. 4.17. If the time taken for the charges to move from the edge of the depletion region ($x = x_n$) to the n-contact ($x = W_n + x_n$) is τ, then the total charge which reaches the contact in time τ is Q_p. Therefore the charge reaching the contact per unit time, which is the current I, is equal to (Q_p / τ). Comparing with eqn.(4.79), we see that $\tau = \tau_t$. So τ_t can be considered to be the time required by the carriers to cross the n-region, and hence the name.

The small signal equivalent circuit of a diode is shown in Fig.4.18. In this figure, the circuit components are G, C_J and C_d given by equations (4.75), (4.31) and (4.74) or (4.80) respectively.

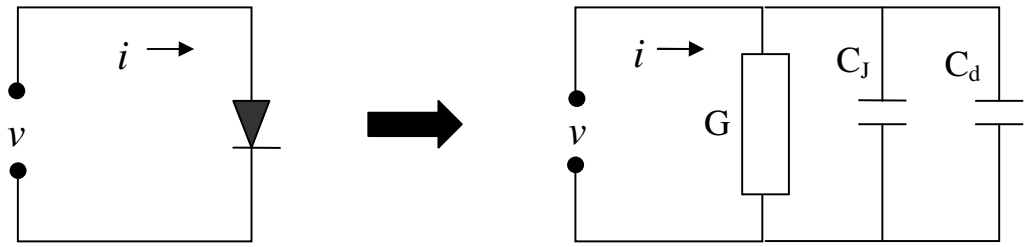


Fig.4.18 The small-signal equivalent circuit of a p-n junction diode

4.5 Breakdown Mechanisms

When a sufficiently large reverse voltage is applied to the diode, the p-n junction 'breaks down' and conducts a large current. The breakdown process is not inherently destructive, i.e. when the applied voltage is reduced the diode again behaves normally. However, the maximum current must be limited by connecting an external resistance (R) as shown in Fig. 4.19(a) to avoid excessive heating. If the breakdown voltage of the diode is V_{BR} as shown in Fig. 4.19(b), then the maximum reverse current that can flow is

$$I_R = \frac{V - V_{BR}}{R} \quad \ldots\ldots(4.81)$$

It must be ensured that the $I_R V_{BR}$ product is less than the maximum power rating of the diode to avoid burn-up. This reverse breakdown encountered in a p-n junction can occur by two mechanisms, each of which needs a critical electric field in the space charge region. One is called **Zener Breakdown,** which usually occurs at low reverse voltages (a few volts). The other mechanism called **Avalanche Breakdown** is responsible for breakdown occurring at higher voltages.

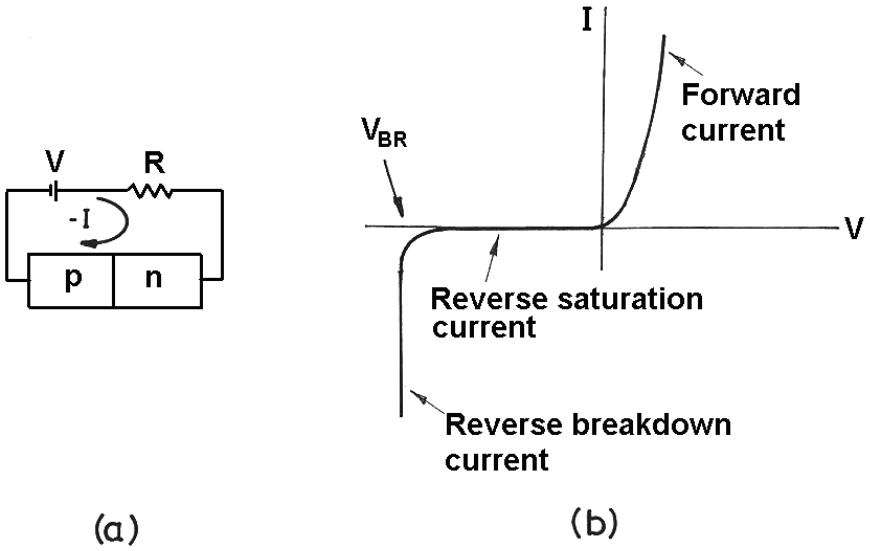


Fig.4.19 (a) A diode circuit with resistive load; (b) the diode characteristics showing the breakdown region

4.5.1 Zener Breakdown

When two heavily doped p and n regions form a junction, then on application of reverse bias, the valence band of the p^+ region can be aligned opposite to the conduction band of the n^+ region as shown in Fig.4.20. In other words, the conduction and valence band edges on either side of the junction may cross even at relatively low voltages. So on the p-side, there is a large density of filled states and on the n-side there are empty states in the conduction band at the same energy. If the barrier separating the p and n regions is narrow (the width of the space charge layer is small), 'tunneling' of electrons can occur from p-side to n-side causing an appreciable reverse current to flow. This is called the **Zener effect**. Since the tunneling probability depends on the width of the barrier, it is important that the barrier is abrupt and doping concentrations on both sides are high ($>5x10^{17}/cm^3$) for the Zener effect to take place.

The Zener effect can be thought of as field ionization of the host atoms at the junction. The applied reverse bias across a very narrow depletion region results in a large electric field inside the space charge region. At the **critical field strength**, bonds are broken as the valence electrons forming the covalent bonds are torn out by the field. The electrons and holes, which are created by this process, move in opposite directions under the influence of the strong electric field in the space charge region, resulting in a large increase in current. The critical electric field for Zener breakdown is of the order of 10^6V/cm. Diodes, which are deliberately designed for a particular Zener breakdown voltage, are often referred to as **Zener diodes**.

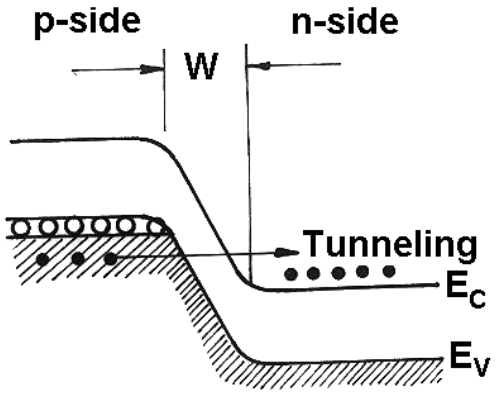


Fig.4.20 Band diagram of a reverse biased p^+-n^+ junction diode showing the Zener effect

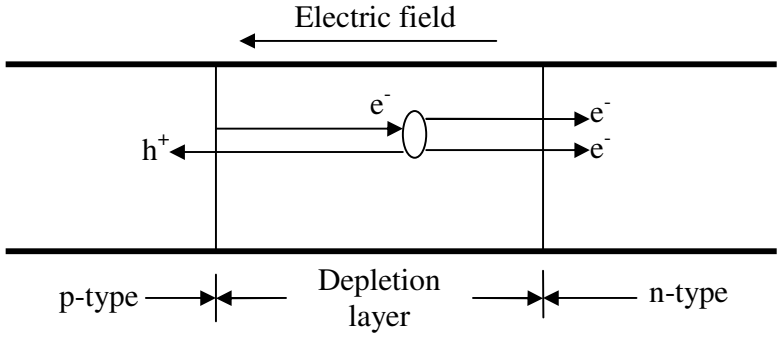


Fig. 4.21 Carrier multiplication in the depletion region due to impact ionization

4.5.2 Avalanche Breakdown

For lightly doped junctions, electron tunneling is negligible. Instead, the breakdown in such cases involves **impact ionization** of atoms by energetic carriers. For example, if the electric field ε inside the depletion region is large, an electron entering this region from the p-side may acquire sufficiently high energy to cause an ionizing collision with a lattice atom generating an electron-hole pair (EHP). A single such event results in **carrier multiplication**; the original electron as well as the generated (secondary) electron are swept to the n-side, while the generated hole is swept to the p-side as shown in Fig.4.21. The degree of multiplication can become very high if the carriers generated within the depletion region also have ionizing collisions with the lattice. That is to say, an incoming carrier undergoes ionizing collision producing an EHP; each of these carriers creates a new EHP; again each new carrier produces an EHP and the process continues. This is called an **avalanche process** since each incoming carrier can create a large number of new carriers.

We shall now make an approximate analysis of this avalanche multiplication of carriers. Let us assume that a carrier (electron or hole), while being accelerated through a depletion region of width W, has a probability P of creating an EHP by undergoing ionizing collision with the lattice. Then for n_{in} incoming electrons entering the depletion region from the p-side, Pn_{in} secondary EHPs will be generated. Now these generated electrons move to the n-side while the generated holes travel to the p-side under the electric field. As the total distance traversed by this EHP is still W, they will in turn generate new EHPs with the same probability and consequently there will be $(Pn_{in})P$

tertiary electrons. Summing up, assuming no recombination, the total number of electrons coming out of the depletion region on the n-side is obtained as

$$n_{out} = n_{in}(1 + P + P^2 + P^3 + ...) \ldots\ldots(4.82)$$

. So, the multiplication factor is given by

$$M = \frac{n_{out}}{n_{in}} = (1 + P + P^2 + P^3 + ...) = \frac{1}{1 - P} \ldots\ldots\ldots(4.83)$$

The probability of an ionizing collision as a carrier travels through the depletion region can also be expressed as

$$P = \int_0^W \alpha dx \ldots\ldots..(4.84)$$

where α is the ionization co-efficient. For the avalanche process to be self-sustaining, M should be infinite, i.e.

$$P = \int_0^W \alpha dx = 1 \ldots\ldots..(4.85)$$

Now, α is a function of the electric field and can be expressed as

$$\alpha = \alpha_o \exp\left[-\left(\frac{b}{\varepsilon}\right)^m\right] \ldots\ldots(4.86)$$

where the constants α_o, b and m are properties of the particular semiconductor.

It must be pointed out that the above analysis is oversimplified and actually the ionization probability is related to the junction parameters in a much more complicated manner. Qualitatively, we can expect the ionization probability to increase with increasing electric field and therefore on the applied reverse bias. A widely used empirical relation between the multiplication factor M and the applied reverse bias voltage (V_r) near breakdown is given by

$$M = \frac{1}{\left[1-\left(\frac{V_r}{V_{BR}}\right)^n\right]} \quad \text{.....(4.87)}$$

where n varies between 3 and 6 depending on the semiconductor material. In general, the critical reverse voltage for breakdown increases with increasing values of bandgap, since more energy is required to cause ionization.

From the breakdown conditions described above and the field dependence of the ionization co-efficient, the critical electric field (ε_c) at which the avalanche process takes place and breakdown occurs, may be calculated as shown in Fig. 4.22. As can be seen from this figure, the value of the critical electric field is only weakly dependent on the doping concentration till tunneling becomes the dominant breakdown mechanism. Now from eqn. (4.26a), noting that $|\varepsilon_m| = \varepsilon_c$ when $V = -V_{BR}$, the breakdown voltage for one-sided abrupt p^+n junction is given by

$$V_{BR} = \frac{\varepsilon_c W}{2} = \frac{\varepsilon_s \varepsilon_c^2}{2qN_D} \quad \text{.....(4.88)}$$

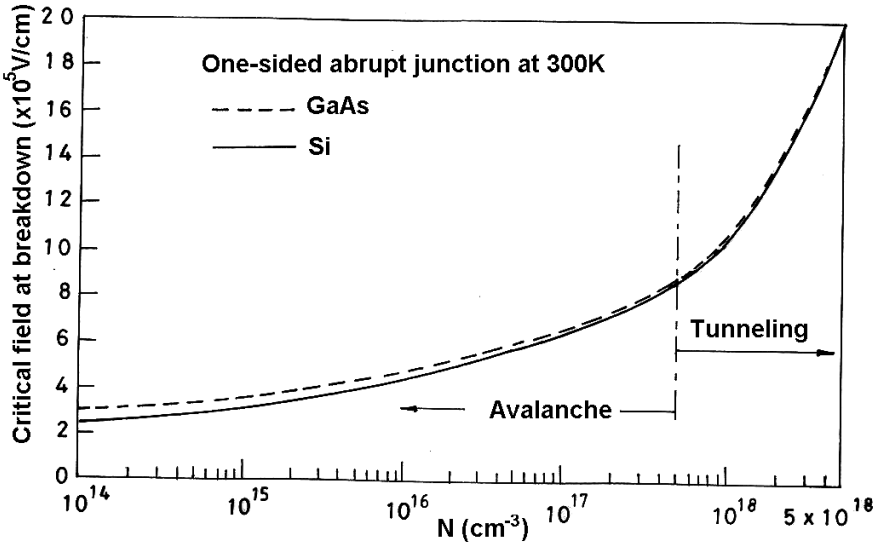


Fig.4.22 Critical field at breakdown versus doping concentration in Si and GaAs[2]

From eqn.(4.88), we see that for junctions where breakdown takes place due to the avalanche process, the breakdown voltage increases with reduction in the doping concentration. The avalanche breakdown voltage is also found to increase with temperature. This is because as the temperature increases, the scattering of the carriers increases. This energy loss obviously does not contribute to the ionization process. That means a larger voltage now has to be applied across the junction to achieve the condition for M to become infinite.

Example 4.6: In a p^+nn^+ junction diode, the doping concentration of the n-region is $2x10^{15}cm^{-3}$. If the critical field at avalanche breakdown is $2x10^5$ V/cm, find out

(a) The breakdown voltage if the width of the n-region is 10 μm

(b) The breakdown voltage if the width of the n-region is 1 μm

(c) Sketch the electric field distribution for both cases.

Solution: (a) For this structure, the depletion region exists essentially only in the n-region. At breakdown the peak electric field is $|\varepsilon_m| = \varepsilon_c = 2x10^5$ V/cm. Substituting this value in eqn.(4.13) and assuming $x_n = W$, we get

$$W = \frac{2x10^5 \, x11.9x8.854x10^{-14}}{1.6x10^{-19} \, x2x10^{15}} cm = 6.585\mu m.$$

The breakdown voltage is now obtained using eqn. (4.88) as

$$V_{BR} = \frac{2x10^5 \, x6.585x10^{-4}}{2} = 65.85V$$

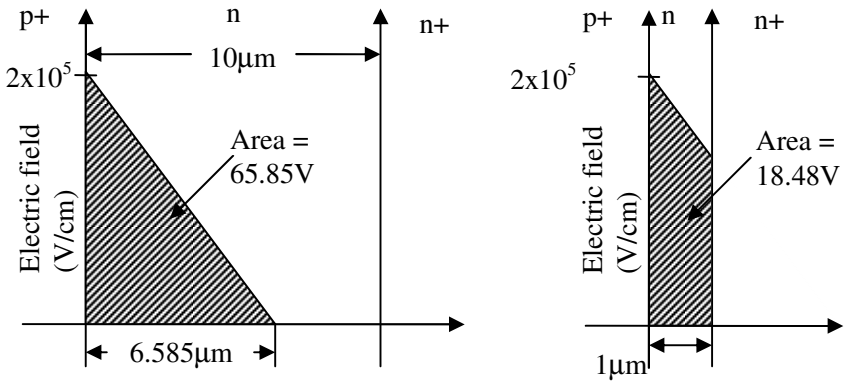
(b) Since, the width of the n-region is now 1 μm, the depletion region width is virtually restricted to 1μm. The extension of depletion region in the heavily doped p^+ and n^+ regions will be negligible and hence can be ignored. As the slope of the electric field

distribution in n-region $\frac{d\varepsilon}{dx} = \frac{qN_D}{\varepsilon_s} = \frac{1.6x10^{-19}\,x2x10^{15}}{11.9x8.854x10^{-14}} = 3.037x10^8\,V/cm^2$ is a constant (depends only on the doping concentration of the n-region), the electric field now takes a trapezoidal shape (instead of the triangular shape as in case a) as shown. From this figure, we calculate the value of the electric field at the nn^+ junction at breakdown as

$$\varepsilon_A = 2x10^5 - (3.037x10^8\,x10^{-4}) = 1.696x10^5\,V/cm.$$

The breakdown voltage can now be calculated as the area under the trapezoid as

$$V_{BR} = \frac{(2+1.696)x10^5}{2}\,x10^{-4} = 18.48V$$



4.6 Fabrication of discrete planar p-n junction diodes

In the previous sections, we have discussed various aspects of the p-n junction diode. Let us now take a look at the fabrication of this device. The basic steps involved in the fabrication of a p-n junction diode using planar technology are outlined in Fig. 4.23. The starting material is a p-type silicon substrate.

Step 1: The wafer is now cleaned to remove organic and/or metallic impurities as well as any trace of native oxide that may be present on the surface. Thermal oxidation is then carried out in order to grow a layer of SiO_2 (500-800 nm) on both top and bottom surface (Fig 4.23a).

Step 2: Next, photoresist is spin-coated and photolithography is carried out in order to realise a mask pattern on the top surface. The bottom surface is fully coated with photoresist (Fig. 4.23b).

Step 3: The sample is now placed in a solution containing HF. HF etches SiO_2 through the openings in the photoresist on the top surface. However, SiO_2 is not affected in the regions protected by photoresist. Photoresist is then removed from everywhere in an organic solvent (acetone in case of positive photoresist) and the structure has "oxide windows" in the top surface as shown in Fig. 4.23c.

Step 4: Phosphorus diffusion is next carried out to produce n^+-regions in the windows. As the diffusion coefficient of phosphorus in SiO_2 is small, phosphorus will only diffuse through the windows, i.e. in the regions no longer protected by the oxide. The p-n junction thus realised is shown in Fig. 4.23d.

Step 5: Oxide is now etched from the back surface of the sample. Aluminium is then deposited on both top and bottom surface of the sample by vacuum evaporation (Fig.4.23e).

Step6: Another photolithography step is next carried out (analogous to step 3) to define with photoresist the areas in which the metal is to be retained. The sample is now placed in an acid solution to remove metal from undesired regions. Photoresist is next removed from everywhere and the sample containing many p-n junction diodes is realized as

shown in Fig. 4.23f. Please note that in this case, all the diodes share the same cathode and have a common cathode contact at the bottom surface. The sample can now be diced into small pieces in order to obtain individual diodes.

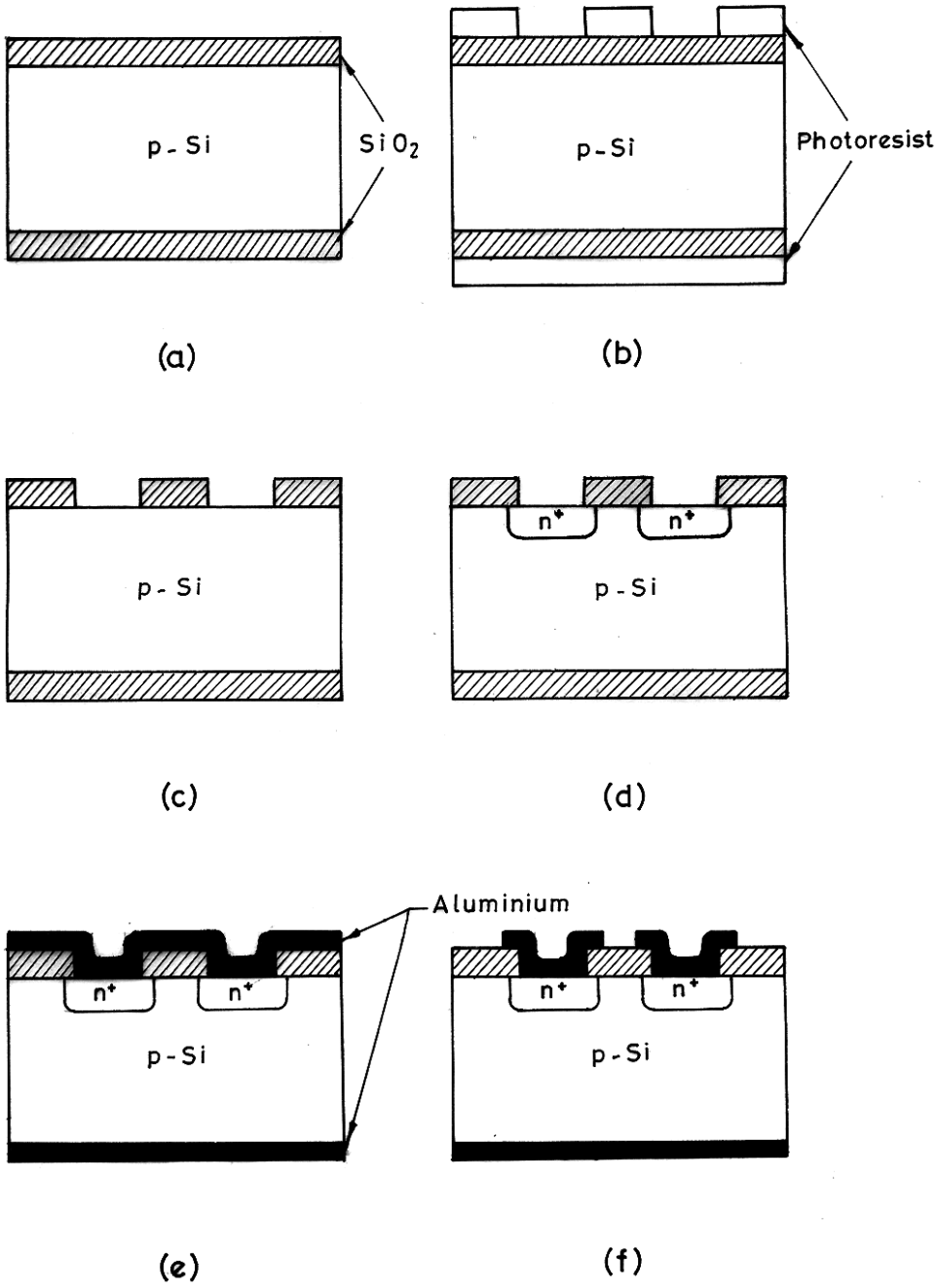


Fig.4.23 Process steps to fabricate p-n junction diodes

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3. B.G. Streetman and S. Banerjee, Solid State Electronic Devices, 5th ed., Prentice Hall, New Jersey, 2000.
4. M.S. Tyagi, Introduction to Semiconductor Materials and Devices, John Wiley & Sons, New York, 1991.

PROBLEMS AND QUESTIONS

1. An abrupt silicon p-n junction has $N_A = 10^{17}$ cm^{-3} on one side and $N_D = 10^{15}$ cm^{-3} on the other. Assuming complete ionisation, calculate the Fermi level positions at 300K in the p and n regions. Also draw the equilibrium band diagram for the junction and determine the contact potential V_{bi} from the diagram.

2. An abrupt p-n junction in silicon is doped with $N_D = 10^{15}$ cm^{-3} on the n-side and $N_A = 4\text{x}10^{20}$ cm^{-3} on the p-side. At room temperature, calculate
   (a) the built-in potential,
   (b) The depletion layer width and the maximum field at zero bias.
   (c) The depletion layer width and the maximum field at a reverse bias of 5V
   (d) The depletion layer width and the maximum field at a forward bias of 0.5V

3. The peak electric field at the junction of a silicon p^+n junction of arbitrary n-doping profile is $1.5\text{x}10^5$ volts/cm. Determine
   (a) The total charge per unit area in the depletion region on the n-side.
   (b) The depletion layer width on the p^+ side of the junction if the doping on this side is uniform and equal to 10^{19}cm^{-3}.
   (c) The voltage across the depletion layer in the p^+ region.

4. In Fig. 4.2(c), the point in the space charge layer at which $E_F = E_i$ is referred to as the intrinsic point. At this point, $n = p = n_i$. Show that the intrinsic point lies on the side of the space charge layer with the lower doping concentration.

5. A p^+n silicon diode is doped with $N_D = 10^{16}$cm^{-3} on the n-side, where D_p=10 cm^2/sec and $\tau_p = 0.1$ μsec. The junction area is 10^{-4} cm^2. Calculate the reverse saturation current and the forward current when V =0.5 volts.

p-region

6. The hole injection efficiency of a junction is defined as I_p/I where I is the total diode current. Assuming that the junction follows the long diode equations, show that $I_p/I = 1/(1+ L_p\sigma_n/L_n\sigma_p)$, where L_n is the diffusion length of electrons in the p-region, L_p is the diffusion length of holes in the n-region, σ_n is the conductivity in the n-region and σ_p is the conductivity of the p-region.

7. Determine N_D and area of a long base abrupt silicon p^+n junction which provides a forward current of 1mA when $V_F = 0.6V$ and has a peak electric field of 20 volts/micron at a reverse bias voltage of 150 volts. Assume that $\tau_p = 0.1\mu s$ on the n-side and $\mu_p = 450\ cm^2/V\text{-sec}$.

8. A p^+n junction with a reverse saturation current I_o is carrying a steady forward current I_1 for $t < 0$. At $t = 0$, the diode current is abruptly increased to I_2. Assuming that the n region is very long and $I_o = 1\mu A$, calculate the stored charge and the diode voltage at $t = 0.5\mu s$ if $I_1 = 1mA$, $I_2 = 10mA$, $\tau_p = 1\mu s$ and $T = 300K$.

9. For a p^+n diode, $\mu_p = 450\ cm^2/V\text{-sec}$ and $\tau_p = 1\mu s$ in the n-region. Calculate the widths of the n-region for which (a) $W_n \leq 0.1L_p$ (i.e. a short-base diode) and (b) $W_n > 4L_p$ (i.e. a long-base diode).

10. How is eqn.(4.62) to be modified for a short-base diode?

11. Show that $I_p(x = x_n) > (Q_p/\tau_p)$ for a diode with a short n-region.